2.5.8 Suppose that a sequence $\{s_n\}$ of positive numbers satisfies the condition $s_{n+1} > \alpha s_n$ for all n where $\alpha > 1$. Show that $s_n \to \infty$.

2.5.9 The sequence $s_n = (-1)^n$ does not diverge to ∞. For what values of M is it nonetheless true that there is an integer N so that $s_n > M$ whenever $n \geq N$?

2.5.10 Show that the sequence

$$n^p + \alpha_1 n^{p-1} + \alpha_2 n^{p-2} + \cdots + \alpha_p$$

diverges to ∞, where here p is a positive integer and α_1, $\alpha_2, \ldots, \alpha_p$ are real numbers (positive or negative).

2.6 Boundedness Properties of Limits

A sequence is said to be *bounded* if its range is a bounded set. Thus a sequence $\{s_n\}$ is bounded if there is a number M so that every term in the sequence satisfies

$$|s_n| \leq M.$$

For such a sequence, every term belongs to the interval $[-M, M]$.

It is fairly evident that a sequence that is not bounded could not converge. This is important enough to state and prove as a theorem.

Theorem 2.11: *Every convergent sequence is bounded.*

Proof. Suppose that $s_n \to L$. Then for every number $\varepsilon > 0$ there is an integer N so that

$$|s_n - L| < \varepsilon$$

whenever $n \geq N$. In particular we could take just one value of ε, say $\varepsilon = 1$, and find a number N so that

$$|s_n - L| < 1$$

whenever $n \geq N$. From this we see that

$$|s_n| = |s_n - L + L| \leq |s_n - L| + |L| < |L| + 1$$

for all $n \geq N$. This number $|L| + 1$ would be an upper bound for all the numbers $|s_n|$ except that we have no indication of the values for $|s_1|$, $|s_2|$, ..., $|s_{N-1}|$.

Thus if we write

$$M = \max\{|s_1|, |s_2|, \ldots, |s_{N-1}|, |L| + 1\}$$

we must have

$$|s_n| \leq M$$

for *every* value of n. This is an upper bound, proving the theorem. ■

As a consequence of this theorem we can conclude that an unbounded sequence must diverge. Thus, even though it is a rather crude test, we can prove the divergence of a sequence if we are able somehow to show that it is unbounded. The next example illustrates this technique.

Example 2.12: We shall show that the sequence

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$$

diverges. The easiest proof of this is to show that it is unbounded and hence, by Theorem 2.11, could not converge.

We watch only at the steps 1, 2, 4, 8, ... and make a rough lower estimate of s_1, s_2, s_4, s_8, ... in order to show that there can be no bound on the sequence. After a bit of arithmetic we see that

$$\begin{aligned} s_1 &= 1 \\ s_2 &= 1 + \frac{1}{2} \\ s_4 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) > 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) \\ s_8 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\ &\geq 1 + \frac{1}{2} + 2\left(\frac{1}{4}\right) + 4\left(\frac{1}{8}\right) \end{aligned}$$

and, in general, that

$$s_{2^n} \geq 1 + n/2$$

for all $n = 0, 1, 2, \ldots$. Thus the sequence is not bounded and so must diverge. ◀

Example 2.13: As a variant of the sequence of the preceding example consider the sequence

$$t_n = 1 + \frac{1}{2^p} + \frac{1}{3^p} + \frac{1}{4^p} + \cdots + \frac{1}{n^p}$$

where p is any positive real number. The case $p = 1$ we have just found diverges.

For $p < 1$ the sequence is larger than it is for $p = 1$ and so the case is even stronger for divergence. For $p > 1$ the sequence is smaller and we cannot see immediately whether it is bounded or unbounded; in fact, with some effort we can show that such a sequence is bounded. What can we conclude? Nothing yet. An unbounded sequence diverges. A bounded sequence may converge or diverge. ◀

Exercises

2.6.1 Which statements are true?

(a) If $\{s_n\}$ is unbounded then it is true that either $\lim_{n\to\infty} s_n = \infty$ or else $\lim_{n\to\infty} s_n = -\infty$.

(b) If $\{s_n\}$ is unbounded then $\lim_{n\to\infty} |s_n| = \infty$.

(c) If $\{s_n\}$ and $\{t_n\}$ are both bounded then so is $\{s_n + t_n\}$.

(d) If $\{s_n\}$ and $\{t_n\}$ are both unbounded then so is $\{s_n + t_n\}$.

(e) If $\{s_n\}$ and $\{t_n\}$ are both bounded then so is $\{s_n t_n\}$.

(f) If $\{s_n\}$ and $\{t_n\}$ are both unbounded then so is $\{s_n t_n\}$.

(g) If $\{s_n\}$ is bounded then so is $\{1/s_n\}$.

(h) If $\{s_n\}$ is unbounded then $\{1/s_n\}$ is bounded.

2.6.2 If $\{s_n\}$ is bounded prove that $\{s_n/n\}$ is convergent.

2.6.3 State the converse of Theorem 2.11. Is it true?

2.6.4 State the contrapositive of Theorem 2.11. Is it true?

2.6.5 Suppose that $\{s_n\}$ is a sequence of positive numbers converging to a positive limit. Show that there is a positive number c so that $s_n > c$ for all n.
See Note 19

2.6.6 As a computer experiment compute the values of the sequence

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$$

for large values of n. Is there any indication in the numbers that you see that this sequence fails to converge or must be unbounded?

2.7 Algebra of Limits

Sequences can be combined by the usual arithmetic operations (addition, subtraction, multiplication, and division). Indeed most sequences we are likely to encounter can be seen to be composed of simpler sequences combined together in this way.

In Example 2.7 we suggested that the computations

$$\lim_{n\to\infty} \frac{n^2}{2n^2+1} = \lim_{n\to\infty} \frac{1}{2+1/n^2} = \frac{1}{\lim_{n\to\infty}(2+1/n^2)}$$

$$= \frac{1}{2+\lim_{n\to\infty} 1/n^2} = \frac{1}{2}$$

could be justified. Note how this sequence has been obtained from simpler ones by ordinary processes of arithmetic. To justify such a method we need to investigate how the limit operation is influenced by algebraic operations.

Suppose that

$$s_n \to S \text{ and } t_n \to T.$$

Then we would expect

$$Cs_n \to CS$$

$$s_n + t_n \to S + T$$

$$s_n - t_n \to S - T$$

$$s_n t_n \to ST$$

and

$$s_n/t_n \to S/T.$$

Each of these statements must be justified, however, solely on the basis of the definition of convergence, not on intuitive feelings that this should be the case. Thus we need to develop what could be called the "algebra of limits."

Theorem 2.14 (Multiples of Limits) *Suppose that $\{s_n\}$ is a convergent sequence and C a real number. Then*

$$\lim_{n\to\infty} Cs_n = C\left(\lim_{n\to\infty} s_n\right).$$

Proof. Let $S = \lim_{n\to\infty} s_n$. In order to prove that $\lim_{n\to\infty} Cs_n = CS$ we need to prove that, no matter what positive number ε is given, we can find an integer N so that, for all $n \geq N$,

$$|Cs_n - CS| < \varepsilon.$$

Note that

$$|Cs_n - CS| = |C|\,|s_n - S|$$

by properties of absolute values. This gives us our clue.

Suppose first that $C \neq 0$ and let $\varepsilon > 0$. Choose N so that

$$|s_n - S| < \varepsilon/|C|$$

if $n \geq N$. Then if $n \geq N$ we must have

$$|Cs_n - CS| = |C|\,|s_n - S| < |C|\,(\varepsilon/|C|) = \varepsilon.$$

This is precisely the statement that

$$\lim_{n\to\infty} Cs_n = CS$$

and the theorem is proved in the case $C \neq 0$. The case $C = 0$ is obvious. (Now we should probably delete our first paragraph since it does not contribute to the proof; it only serves to motivate us in finding the correct proof.) ■

Theorem 2.15 (Sums/Differences of Limits) *Suppose that the sequences $\{s_n\}$ and $\{t_n\}$ are convergent. Then*

$$\lim_{n\to\infty} (s_n + t_n) = \lim_{n\to\infty} s_n + \lim_{n\to\infty} t_n$$

and

$$\lim_{n\to\infty} (s_n - t_n) = \lim_{n\to\infty} s_n - \lim_{n\to\infty} t_n.$$

Proof. Let $S = \lim_{n\to\infty} s_n$ and $T = \lim_{n\to\infty} t_n$. In order to prove that

$$\lim_{n\to\infty} (s_n + t_n) = S + T$$

we need to prove that no matter what positive number ε is given we can find an integer N so that

$$|(s_n + t_n) - (S + T)| < \varepsilon$$

if $n \geq N$. Note that

$$|(s_n + t_n) - (S + T)| \leq |s_n - S| + |t_n - T|$$

by the triangle inequality. Thus we can make this expression smaller than ε by making each of the two expressions on the right smaller than $\varepsilon/2$. This provides the method.

Suppose that $\varepsilon > 0$. Choose N_1 so that

$$|s_n - S| < \varepsilon/2$$

if $n \geq N_1$ and also choose N_2 so that

$$|t_n - T| < \varepsilon/2$$

if $n \geq N_2$. Then if n is greater than both N_1 and N_2 both of these inequalities will be true. Set

$$N = \max\{N_1, N_2\}$$

and note that if $n \geq N$ we must have

$$|(s_n + t_n) - (S+T)| \leq |s_n - S| + |t_n - T| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

This is precisely the statement that

$$\lim_{n\to\infty}(s_n + t_n) = S + T$$

and the first statement of the theorem is proved. The second statement is similar and is left as an exercise. (Once again, for a more formal presentation, we would delete the first paragraph.) ■

Theorem 2.16 (Products of Limits) *Suppose that $\{s_n\}$ and $\{t_n\}$ are convergent sequences. Then*

$$\lim_{n\to\infty}(s_n t_n) = \left(\lim_{n\to\infty} s_n\right)\left(\lim_{n\to\infty} t_n\right).$$

Proof. Let $S = \lim_{n\to\infty} s_n$ and $T = \lim_{n\to\infty} t_n$. In order to prove that $\lim_{n\to\infty}(s_n t_n) = ST$ we need to prove that no matter what positive number ε is given we can find an integer N so that, for all $n \geq N$,

$$|s_n t_n - ST| < \varepsilon.$$

It takes some experimentation with different ways of writing this to find the most useful version. Here is an inequality that offers the best approach:

$$\begin{aligned} |s_n t_n - ST| &= |s_n(t_n - T) + s_n T - ST| \\ &\leq |s_n|\,|t_n - T| + |T|\,|s_n - S|. \end{aligned} \tag{1}$$

We can control the size of $|s_n - S|$ and $|t_n - T|$, T is constant, and $|s_n|$ cannot be too big. To control the size of $|s_n|$ we need to recall that convergent sequences are bounded (Theorem 2.11) and get a bound from there. With these preliminaries explained the rest of the proof should seem less mysterious. (Now this paragraph can be deleted for a more formal presentation.)

Suppose that $\varepsilon > 0$. Since $\{s_n\}$ converges it is bounded and hence, by Theorem 2.11, there is a positive number M so that $|s_n| \leq M$ for all n. Choose N_1 so that

$$|s_n - S| < \frac{\varepsilon}{2|T| + 1}$$

if $n \geq N_1$. [We did not use $\varepsilon/(2T)$ since there is a possibility that $T = 0$.] Also, choose N_2 so that

$$|t_n - T| < \frac{\varepsilon}{2M}$$

if $n \geq N_2$. Set $N = \max\{N_1, N_2\}$ and note that if $n \geq N$ we must have

$$|s_n t_n - ST| \leq |s_n|\,|t_n - T| + |T|\,|s_n - S|$$
$$\leq M\left(\frac{\varepsilon}{2M}\right) + |T|\left(\frac{\varepsilon}{2|T|+1}\right) < \varepsilon.$$

This is precisely the statement that

$$\lim_{n\to\infty} s_n t_n = ST$$

and the theorem is proved. ■

Theorem 2.17 (Quotients of Limits) *Suppose that $\{s_n\}$ and $\{t_n\}$ are convergent sequences. Suppose further that $t_n \neq 0$ for all n and that the limit*

$$\lim_{n\to\infty} t_n \neq 0.$$

Then

$$\lim_{n\to\infty}\left(\frac{s_n}{t_n}\right) = \frac{\lim_{n\to\infty} s_n}{\lim_{n\to\infty} t_n}.$$

Proof. Rather than prove the theorem at once as it stands let us prove just a special case of the theorem, namely that

$$\lim_{n\to\infty}\left(\frac{1}{t_n}\right) = \frac{1}{\lim_{n\to\infty} t_n}.$$

Let $T = \lim_{n\to\infty} t_n$. We need to show that no matter what positive number ε is given we can find an integer N so that

$$\left|\frac{1}{t_n} - \frac{1}{T}\right| < \varepsilon$$

if $n \geq N$. To work with this inequality requires us to consider

$$\left|\frac{1}{t_n} - \frac{1}{T}\right| = \frac{|t_n - T|}{|t_n|\,|T|}.$$

It is only the $|t_n|$ in the denominator that offers any trouble since if it is too small we cannot control the size of the fraction. This explains the first step in the proof that we now give, which otherwise might have seemed strange.

Suppose that $\varepsilon > 0$. Choose N_1 so that

$$|t_n - T| < |T|/2$$

if $n \geq N_1$ and also choose N_2 so that

$$|t_n - T| < \varepsilon|T|^2/2$$

if $n \geq N_2$. From the first inequality we see that

$$|T| - |t_n| \leq |T - t_n| < |T|/2$$

and so

$$|t_n| \geq |T|/2$$

if $n \geq N_1$. Set $N = \max\{N_1, N_2\}$ and note that if $n \geq N$ we must have

$$\left|\frac{1}{t_n} - \frac{1}{T}\right| = \frac{|t_n - T|}{|t_n|\,|T|}$$

$$< \frac{\varepsilon|T|^2/2}{|T|^2/2} = \varepsilon.$$

This is precisely the statement that $\lim_{n\to\infty}(1/t_n) = 1/T$.

We now complete the proof of the theorem by applying the product theorem along with what we have just proved to obtain

$$\lim_{n\to\infty}\left(\frac{s_n}{t_n}\right) = \left(\lim_{n\to\infty} s_n\right)\left(\lim_{n\to\infty}\frac{1}{t_n}\right) = \frac{\lim_{n\to\infty} s_n}{\lim_{n\to\infty} t_n}$$

as required. ■

Exercises

2.7.1 By imitating the proof given for the first part of Theorem 2.15 show that

$$\lim_{n\to\infty}(s_n - t_n) = \lim_{n\to\infty} s_n - \lim_{n\to\infty} t_n.$$

2.7.2 Show that $\lim_{n\to\infty}(s_n)^2 = (\lim_{n\to\infty} s_n)^2$ using the theorem on products and also directly from the definition of limit.

2.7.3 Explain which theorems are needed to justify the computation of the limit

$$\lim_{n\to\infty}\frac{n^2}{2n^2+1}$$

that introduced this section.

2.7.4 Prove Theorem 2.16 but verifying and using the inequality

$$|s_n t_n - ST| \leq |(s_n - S)(t_n - T)| + |S(t_n - T)| + |T(s_n - S)|$$

in place of the inequality (1). Which proof do you prefer?

2.7.5 Which statements are true?

(a) If $\{s_n\}$ and $\{t_n\}$ are both divergent then so is $\{s_n + t_n\}$.

(b) If $\{s_n\}$ and $\{t_n\}$ are both divergent then so is $\{s_n t_n\}$.

(c) If $\{s_n\}$ and $\{s_n + t_n\}$ are both convergent then so is $\{t_n\}$.

(d) If $\{s_n\}$ and $\{s_n t_n\}$ are both convergent then so is $\{t_n\}$.

(e) If $\{s_n\}$ is convergent so too is $\{1/s_n\}$.

(f) If $\{s_n\}$ is convergent so too is $\{(s_n)^2\}$.

(g) If $\{(s_n)^2\}$ is convergent so too is $\{s_n\}$.

2.7.6 Note that there are extra hypotheses in the quotient theorem (Theorem 2.17) that were not in the product theorem (Theorem 2.16). Explain why both of these hypotheses are needed.

2.7.7 A careless student gives the following as a proof of Theorem 2.16. Find the flaw:
"Suppose that $\varepsilon > 0$. Choose N_1 so that

$$|s_n - S| < \frac{\varepsilon}{2|T|+1}$$

if $n \geq N_1$ and also choose N_2 so that

$$|t_n - T| < \frac{\varepsilon}{2|s_n|+1}$$

if $n \geq N_2$. If $n \geq N = \max\{N_1, N_2\}$ then

$$|s_n t_n - ST| \leq |s_n|\,|t_n - T| + |T|\,|s_n - S|$$
$$\leq |s_n| \left(\frac{\varepsilon}{2|s_n|+1}\right) + |T| \left(\frac{\varepsilon}{2|T|+1}\right) < \varepsilon.$$

Well, that works!"

2.7.8 Why are Theorems 2.15 and 2.16 no help in dealing with the limits

$$\lim_{n\to\infty} \left(\sqrt{n+1} - \sqrt{n}\right)$$

and

$$\lim_{n\to\infty} \sqrt{n}\left(\sqrt{n+1} - \sqrt{n}\right)?$$

What else can you do?

2.7.9 In calculus courses one learns that a function $f : \mathbb{R} \to \mathbb{R}$ is continuous at y if for every $\varepsilon > 0$ there is a $\delta > 0$ so that $|f(x) - f(y)| < \varepsilon$ for all $|x - y| < \delta$. Show that if f is continuous at y and $s_n \to y$ then $f(s_n) \to f(y)$. Use this to prove that $\lim_{n\to\infty}(s_n)^2 = (\lim_{n\to\infty} s_n)^2$.

2.8 Order Properties of Limits

In the preceding section we discussed the algebraic structure of limits. It is a natural mathematical question to ask how the algebraic operations are preserved under limits. As it happens, these natural mathematical questions usually are important in applications. We have seen that the algebraic properties of limits can be used to great advantage in computations of limits.

There is another aspect of structure of the real number system that plays an equally important role as the algebraic structure and that is the order structure. Does the limit operation preserve that order structure the same way that it preserves the algebraic structure? For example, if

$$s_n \leq t_n$$

for all n, can we conclude that

$$\lim_{n\to\infty} s_n \leq \lim_{n\to\infty} t_n?$$

In this section we solve this problem and several others related to the order structure. These results, too, will prove to be most useful in handling limits.

Theorem 2.18: *Suppose that $\{s_n\}$ and $\{t_n\}$ are convergent sequences and that*

$$s_n \leq t_n$$

for all n. Then

$$\lim_{n\to\infty} s_n \leq \lim_{n\to\infty} t_n.$$

Proof. Let $S = \lim_{n\to\infty} s_n$ and $T = \lim_{n\to\infty} t_n$ and suppose that $\varepsilon > 0$. Choose N_1 so that

$$|s_n - S| < \varepsilon/2$$

if $n \geq N_1$ and also choose N_2 so that

$$|t_n - T| < \varepsilon/2$$

if $n \geq N_2$. Set $N = \max\{N_1, N_2\}$ and note that if $n \geq N$ we must have

$$0 \leq t_n - s_n = T - S + (t_n - T) + (S - s_n) < T - S + \varepsilon/2 + \varepsilon/2.$$

This shows that

$$-\varepsilon < T - S.$$

This statement is true for *any* positive number ε. It would be false if $T - S$ is negative and hence $T - S$ is positive or zero (i.e., $T \geq S$ as required). ■

Note. There is a trap here that many students have fallen into. Since the condition $s_n \leq t_n$ implies

$$\lim_{n\to\infty} s_n \leq \lim_{n\to\infty} t_n$$

would it not follow "similarly" that the condition $s_n < t_n$ implies

$$\lim_{n\to\infty} s_n < \lim_{n\to\infty} t_n?$$

Be careful with this. It is false. See Exercise 2.8.1.

Corollary 2.19: *Suppose that $\{s_n\}$ is a convergent sequence and that*

$$\alpha \leq s_n \leq \beta$$

for all n. Then

$$\alpha \leq \lim_{n\to\infty} s_n \leq \beta.$$

Proof. Consider that the assumption here can be read as $\alpha_n \leq s_n \leq \beta_n$ where $\{\alpha_n\}$ and $\{\beta_n\}$ are constant sequences. Now apply the theorem. ■

Note. Again, don't forget the trap. The condition $\alpha < s_n < \beta$ for all n implies that

$$\alpha \leq \lim_{n\to\infty} s_n \leq \beta.$$

It would not imply that

$$\alpha < \lim_{n\to\infty} s_n < \beta.$$

The Squeeze Theorem The next theorem is another useful variant on these themes. Here an unknown sequence is sandwiched between two convergent sequences, allowing us to conclude that that sequence converges. This theorem is often taught as "the squeeze theorem," which seems a convenient label.

Theorem 2.20 (Squeeze Theorem) *Suppose that $\{s_n\}$ and $\{t_n\}$ are convergent sequences, that*

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} t_n$$

and that

$$s_n \le x_n \le t_n$$

for all n. Then $\{x_n\}$ is also convergent and

$$\lim_{n\to\infty} x_n = \lim_{n\to\infty} s_n = \lim_{n\to\infty} t_n.$$

Proof. Let L be the limit of the two sequences. Choose N_1 so that

$$|s_n - L| < \varepsilon$$

if $n \ge N_1$ and also choose N_2 so that

$$|t_n - L| < \varepsilon$$

if $n \ge N_2$. Set $N = \max\{N_1, N_2\}$. Note that

$$s_n - L \le x_n - L \le t_n - L$$

for all n and so

$$-\varepsilon < s_n - L \le x_n - L \le t_n - L < \varepsilon$$

if $n \ge N$. From this we see that

$$-\varepsilon < x_n - L < \varepsilon$$

or, to put it in a more familiar form,

$$|x_n - L| < \varepsilon$$

proving the statement of the theorem. ■

Example 2.21: Let θ be some real number and consider the computation of

$$\lim_{n\to\infty} \frac{\sin n\theta}{n}.$$

While this might seem hopeless at first sight since the values of $\sin n\theta$ are quite unpredictable, we recall that none of these values lies outside the interval $[-1,1]$. Hence

$$-\frac{1}{n} \le \frac{\sin n\theta}{n} \le \frac{1}{n}.$$

The two outer sequences converge to the same value 0 and so the inside sequence (the "squeezed" one) must converge to 0 as well. ◄

Absolute Values A further theorem on the theme of order structure is often needed. The absolute value, we recall, is defined directly in terms of the order structure. Is absolute value preserved by the limit operation?

Theorem 2.22 (Limits of Absolute Values) *Suppose that $\{s_n\}$ is a convergent sequence. Then the sequence $\{|s_n|\}$ is also a convergent sequence and*

$$\lim_{n\to\infty} |s_n| = \left|\lim_{n\to\infty} s_n\right|.$$

Proof. Let $S = \lim_{n\to\infty} s_n$ and suppose that $\varepsilon > 0$. Choose N so that

$$|s_n - S| < \varepsilon$$

if $n \geq N$. Observe that, because of the triangle inequality, this means that

$$||s_n| - |S|| \leq |s_n - S| < \varepsilon$$

for all $n \geq N$. By definition

$$\lim_{n\to\infty} |s_n| = |S|$$

as required. ■

Maxima and Minima Since maxima and minima can be expressed in terms of absolute values, there is a corollary that is sometimes useful.

Corollary 2.23 (Max/Min of Limits) *Suppose that $\{s_n\}$ and $\{t_n\}$ are convergent sequences. Then the sequences*

$$\{\max\{s_n, t_n\}\} \quad \textit{and} \quad \{\min\{s_n, t_n\}\}$$

are also convergent and

$$\lim_{n\to\infty} \max\{s_n, t_n\} = \max\{\lim_{n\to\infty} s_n, \lim_{n\to\infty} t_n\}$$

and

$$\lim_{n\to\infty} \min\{s_n, t_n\} = \min\{\lim_{n\to\infty} s_n, \lim_{n\to\infty} t_n\}.$$

Proof. The first of these follows from the identity

$$\max\{s_n, t_n\} = \frac{s_n + t_n}{2} + \frac{|s_n - t_n|}{2}$$

and the theorem on limits of sums and the theorem on limits of absolute values. In the same way the second assertion follows from

$$\min\{s_n, t_n\} = \frac{s_n + t_n}{2} - \frac{|s_n - t_n|}{2}.$$

■

Exercises

2.8.1 Show that the condition $s_n < t_n$ does not imply that

$$\lim_{n\to\infty} s_n < \lim_{n\to\infty} t_n.$$

(If the proof of Theorem 2.18 were modified in an attempt to prove this false statement, where would the modifications fail?)

SEE NOTE 20

2.8.2 If $\{s_n\}$ is a sequence all of whose values lie inside an interval $[a,b]$ prove that $\{s_n/n\}$ is convergent.

2.8.3 A careless student gives the following as a proof of the squeeze theorem. Find the flaw:

"If $\lim_{n\to\infty} s_n = \lim_{n\to\infty} t_n = L$, then take limits in the inequality

$$s_n \le x_n \le t_n$$

to get $L \le \lim_{n\to\infty} x_n \le L$. This can only be true if $\lim_{n\to\infty} x_n = L$."

2.8.4 Suppose that $s_n \le t_n$ for all n and that $s_n \to \infty$. What can you conclude?

2.8.5 Suppose that $\lim_{n\to\infty} \frac{s_n}{n} > 0$ Show that $s_n \to \infty$.

2.8.6 Suppose that $\{s_n\}$ and $\{t_n\}$ are sequences of positive numbers, that

$$\lim_{n\to\infty} \frac{s_n}{t_n} = \alpha$$

and that $s_n \to \infty$. What can you conclude?

2.8.7 Suppose that $\{s_n\}$ and $\{t_n\}$ are sequences of positive numbers, that

$$\lim_{n\to\infty} \frac{s_n}{t_n} = \infty$$

and that $t_n \to \infty$. What can you conclude?

2.8.8 Suppose that $\{s_n\}$ and $\{t_n\}$ are sequences of positive numbers, that

$$\lim_{n\to\infty} \frac{s_n}{t_n} = \infty$$

and that $\{s_n\}$ is bounded. What can you conclude?

2.8.9 Let $\{s_n\}$ be a sequence of positive numbers. Show that the condition

$$\lim_{n\to\infty} \frac{s_{n+1}}{s_n} < 1$$

implies that $s_n \to 0$.

SEE NOTE 21

2.8.10 Let $\{s_n\}$ be a sequence of positive numbers. Show that the condition

$$\lim_{n\to\infty} \frac{s_{n+1}}{s_n} > 1$$

implies that $s_n \to \infty$.

SEE NOTE 22

2.9 Monotone Convergence Criterion

In many applications of sequence theory we find that the sequences that arise are going in one direction: The terms steadily get larger or steadily get smaller. The analysis of such sequences is much easier than for general sequences.

Definition 2.24: (Increasing) We say that a sequence $\{s_n\}$ is *increasing* if

$$s_1 < s_2 < s_3 < \cdots < s_n < s_{n+1} < \ldots.$$

Definition 2.25: (Decreasing) We say that a sequence $\{s_n\}$ is *decreasing* if

$$s_1 > s_2 > s_3 > \cdots > s_n > s_{n+1} > \ldots.$$

Often we encounter sequences that "increase" except perhaps occasionally successive values are equal rather than strictly larger. The following language is usually[1] used in this case.

Definition 2.26: (Nondecreasing) We say that a sequence $\{s_n\}$ is *nondecreasing* if

$$s_1 \leq s_2 \leq s_3 \leq \cdots \leq s_n \leq s_{n+1} \leq \ldots.$$

Definition 2.27: (Nonincreasing) We say that a sequence $\{s_n\}$ is *nonincreasing* if

$$s_1 \geq s_2 \geq s_3 \geq \cdots \geq s_n \geq s_{n+1} \geq \ldots.$$

Thus every increasing sequence is also nondecreasing but not conversely. A sequence that has any one of these four properties (increasing, decreasing, nondecreasing, or nonincreasing) is said to be *monotonic*. Monotonic sequences are often easier to deal with than sequences that can go both up and down.

The convergence issue for a monotonic sequence is particularly straightforward. We can imagine that an increasing sequence could increase up to some limit, or we could imagine that it could increase indefinitely and diverge to $+\infty$. It is impossible to imagine a third possibility. We express this as a theorem that will become our primary theoretical tool in investigating convergence of sequences.

Theorem 2.28 (Monotone Convergence Theorem) *Suppose that $\{s_n\}$ is a monotonic sequence. Then $\{s_n\}$ is convergent if and only if $\{s_n\}$ is bounded. More specifically,*

1. *If $\{s_n\}$ is nondecreasing then either $\{s_n\}$ is bounded and converges to $\sup\{s_n\}$ or else $\{s_n\}$ is unbounded and $s_n \to \infty$.*

2. *If $\{s_n\}$ is nonincreasing then either $\{s_n\}$ is bounded and converges to $\inf\{s_n\}$ or else $\{s_n\}$ is unbounded and $s_n \to -\infty$.*

[1] In some texts you will find that a nondecreasing sequence is said to be increasing and an increasing sequence is said to be *strictly* increasing. The way in which we intend these terms should be clear and intuitive. If your monthly salary occasionally rises but sometimes stays the same you would not likely say that it is increasing. You might, however, say "at least it never decreases" (i.e., it is nondecreasing).

Proof. If the sequence is unbounded then it diverges. This is true for any sequence, not merely monotonic sequences.

Thus the proof is complete if we can show that for any bounded monotonic sequence $\{s_n\}$ the limit is $\sup\{s_n\}$ in case the sequence is nondecreasing, or it is $\inf\{s_n\}$ in case the sequence is nonincreasing. Let us prove the first of these cases.

Let $\{s_n\}$ be assumed to be nondecreasing and bounded, and let

$$L = \sup\{s_n\}.$$

Then $s_n \leq L$ for all n and if $\beta < L$ there must be some term s_m say, with $s_m > \beta$. Let $\varepsilon > 0$. We know that there is an m so that

$$s_n \geq s_m > L - \varepsilon$$

for all $n \geq m$. But we already know that every term $s_n \leq L$. Putting these together we have that

$$L - \varepsilon < s_n \leq L < L + \varepsilon$$

or

$$|s_n - L| < \varepsilon$$

for all $n \geq m$. By definition then $s_n \to L$ as required. ■

How would we normally apply this theorem? Suppose a sequence $\{s_n\}$ were given that we recognize as increasing (or maybe just nondecreasing). Then to establish that $\{s_n\}$ converges we need only show that the sequence is bounded above, that is, we need to find just one number M with

$$s_n \leq M$$

for all n. Any crude upper estimate would verify convergence.

Example 2.29: Let us show that the sequence $s_n = 1/\sqrt{n}$ converges. This sequence is evidently decreasing. Can we find a lower bound? Yes, all of the terms are positive so that 0 is a lower bound. Consequently, the sequence must converge. If we wish to show that

$$\lim_{n\to\infty} \frac{1}{\sqrt{n}} = 0$$

we need to do more. But to conclude convergence we needed only to make a crude estimate on how low the terms might go. ◄

Example 2.30: Let us examine the sequence

$$s_n = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}.$$

This sequence is evidently increasing. Can we find an upper bound? If we can then the series does converge. If we cannot then the series diverges. We have already (earlier) checked this sequence. It is unbounded and so $\lim_{n\to\infty} s_n = \infty$. ◄

Example 2.31: Let us examine the sequence

$$\sqrt{2},\ \sqrt{2+\sqrt{2}},\ \sqrt{2+\sqrt{2+\sqrt{2}}},\ \sqrt{2+\sqrt{2+\sqrt{2+\sqrt{2}}}},\ldots.$$

Handling such a sequence directly by the limit definition seems quite impossible. This sequence can be defined recursively by

$$x_1 = \sqrt{2} \quad x_n = \sqrt{2+x_{n-1}}.$$

The computation of a few terms suggests that the sequence is increasing and so should be accessible by the methods of this section.

We prove this by induction. That $x_1 < x_2$ is just an easy computation (do it). Let us suppose that $x_{n-1} < x_n$ for some n and show that it must follow that $x_n < x_{n+1}$. But

$$x_n = \sqrt{2+x_{n-1}} < \sqrt{2+x_n} = x_{n+1}$$

where the middle step is the induction hypothesis (i.e., that $x_{n-1} < x_n$). It follows by induction that the sequence is increasing.

Now we show inductively that the sequence is bounded above. Any crude upper bound will suffice. It is clear that $x_1 < 10$. If $x_{n-1} < 10$ then

$$x_n = \sqrt{2+x_{n-1}} < \sqrt{2+10} < 10$$

and so it follows, again by induction, that all terms of the sequence are smaller than 10. We conclude from the monotone convergence theorem that this sequence is convergent.

But to what? (Certainly it does not converges to 10 since that estimate was extremely crude.) That is not so easy to sort out, it seems. But perhaps it is, since we know that the sequence converges to something, say L. In the equation

$$(x_n)^2 = 2 + x_{n-1},$$

obtained by squaring the recursion formula given to us, we can take limits as $n \to \infty$. Since $x_n \to L$ so too does $x_{n-1} \to L$ and $(x_n)^2 \to L^2$. Hence

$$L^2 = 2 + L.$$

The only possibilities for L in this quadratic equation are $L = -1$ and $L = 2$. We know the limit L exists and we know that it is either -1 or 2. We can clearly rule out -1 as all of the numbers in our sequence were positive. Hence $x_n \to 2$. ◀

Exercises

2.9.1 Define a sequence $\{s_n\}$ recursively by setting $s_1 = \alpha$ and

$$s_n = \frac{(s_{n-1})^2 + \beta}{2s_{n-1}}$$

where $\alpha, \beta > 0$.

(a) Show that for $n = 1, 2, 3, \ldots$

$$\frac{(s_n - \sqrt{\beta})^2}{2s_n} = s_{n+1} - \beta.$$

(b) Show that $s_n > \sqrt{\beta}$ for all $n = 2, 3, 4, \ldots$ unless $\alpha = \sqrt{\beta}$. What happens if $\alpha = \sqrt{\beta}$?

(c) Show that $s_2 > s_3 > s_4 > \ldots s_n > \ldots$ except in the case $\alpha = \sqrt{\beta}$.

(d) Does this sequence converge? To what?

(e) What is the relation of this sequence to the one introduced in Section 2.1 as Newton's method?

2.9.2 Define a sequence $\{t_n\}$ recursively by setting $t_1 = 1$ and

$$t_n = \sqrt{t_{n-1} + 1}.$$

Does this sequence converge? To what?

2.9.3 Consider the sequence $s_1 = 1$ and $s_n = \frac{2}{s_{n-1}^2}$. We argue that if $s_n \to L$ then $L = \frac{2}{L^2}$ and so $L^3 = 2$ or $L = \sqrt[3]{2}$. Our conclusion is that $\lim_{n\to\infty} s_n = \sqrt[3]{2}$. Do you have any criticisms of this argument?

2.9.4 Does the sequence

$$\frac{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1)}{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n)}$$

converge?

2.9.5 Does the sequence

$$\frac{2 \cdot 4 \cdot 6 \cdot \cdots \cdot (2n) \cdot 1}{1 \cdot 3 \cdot 5 \cdot \cdots \cdot (2n-1) \cdot n^2}$$

converge?

2.9.6 Several nineteenth-century mathematicians used, without proof, a principle in their proofs that has come to be known as the *nested interval property*:

> *Given a sequence of closed intervals*
>
> $$[a_1, b_1] \supset [a_2, b_2] \supset [a_3, b_3] \supset \ldots$$
>
> *arranged so that each interval is a subinterval of the one preceding it and so that the lengths of the intervals shrink to zero, then there is exactly one point that belongs to every interval of the sequence.*

Prove this statement. Would it be true for a descending sequence of open intervals

$$(a_1, b_1) \supset (a_2, b_2) \supset (a_3, b_3) \supset \ldots?$$

2.10 Examples of Limits

The theory of sequence limits has now been developed far enough that we may investigate some interesting limits. Each of the limits in this section has some cultural interest. Most students would be expected to know and recognize these limits as they arise quite routinely. For us they are also an opportunity to show off our methods. Mostly we need to establish inequalities and use some of our theory. We do not need to use an ε, N argument since we now have more subtle and powerful tools at hand.

Example 2.32: (Geometric Progressions) Let r be a real number. What is the limiting behavior of the sequence

$$1, r, r^2, r^3, r^4, \ldots, r^n, \ldots$$

forming a geometric progression? If $r > 1$ then it is not hard to show that

$$r^n \to \infty.$$

If $r \leq -1$ the sequence certainly diverges. If $r = 1$ this is just a constant sequence.

The interesting case is

$$\lim_{n\to\infty} r^n = 0 \quad \text{if } -1 < r < 1.$$

To prove this we shall use an easy inequality. Let $x > 0$ and n an integer. Then, using the binomial theorem (or induction if you prefer), we can show that

$$(1+x)^n > nx.$$

Case (i): Let $0 < r < 1$. Then

$$r = \frac{1}{1+x}$$

(where $x = 1/r - 1 > 0$) and so

$$0 < r^n = \frac{1}{(1+x)^n} < \frac{1}{nx} \to 0$$

as $n \to \infty$. By the squeeze theorem we see that $r^n \to 0$ as required.

Case (ii): If $-1 < r < 0$ then $r = -t$ for $0 < t < 1$. Thus

$$-t^n \leq r^n \leq t^n.$$

By case (i) we know that $t^n \to 0$. By the squeeze theorem we see that $r^n \to 0$ again as required. ◀

Example 2.33: (Roots) An interesting and often useful limit is

$$\lim_{n\to\infty} \sqrt[n]{n} = 1.$$

To show this we once again derive an inequality from the binomial theorem. If $n \geq 2$ and $x > 0$ then

$$(1+x)^n > n(n-1)x^2/2.$$

For $n \geq 2$ write

$$\sqrt[n]{n} = 1 + x_n$$

(where $x_n = \sqrt[n]{n} - 1 > 0$) and so

$$n = (1+x_n)^n > n(n-1)x_n^2/2$$

or

$$0 < x_n^2 < \frac{2}{n-1} \to 0$$

as $n \to \infty$. By the squeeze theorem we see that $x_n \to 0$ and it follows that $\sqrt[n]{n} \to 1$ as required.

As a special case of this example note that

$$\sqrt[n]{C} \to 1$$

as $n \to \infty$ for any positive constant C. This is true because if $C > 1$ then

$$1 < \sqrt[n]{C} < \sqrt[n]{n}$$

for large enough n. By the squeeze theorem this shows that $\sqrt[n]{C} \to 1$. If, however, $0 < C < 1$ then

$$\sqrt[n]{C} = \frac{1}{\sqrt[n]{1/C}} \to 1$$

by the first case since $1/C > 1$. ◀

Example 2.34: (Sums of Geometric Progressions) For all values of x in the interval $(-1,1)$ the limit

$$\lim_{n\to\infty} \left(1+x+x^2+x^3+\cdots+x^n\right) = \frac{1}{1-x}.$$

While at first a surprising result, this is quite evident once we check the identity

$$(1-x)\left(1+x+x^2+x^3+\cdots+x^n\right) = 1-x^{n+1},$$

which just requires a straightforward multiplication. Thus

$$\lim_{n\to\infty} \left(1+x+x^2+x^3+\cdots+x^n\right) = \lim_{n\to\infty} \frac{1-x^{n+1}}{1-x} = \frac{1}{1-x}$$

where we have used the result we proved previously, namely that

$$x^{n+1} \to 0 \quad \text{if } |x| < 1.$$

One special case of this is useful to remember. Set $x = 1/2$. Then

$$\lim_{n\to\infty} \left(1+\frac{1}{2}+\frac{1}{2^2}+\frac{1}{2^3}+\cdots+\frac{1}{2^n}\right) = 2.$$

◀

Example 2.35: (Decimal Expansions) What meaning is assigned to the infinite decimal expansion

$$x = 0.d_1d_2d_3d_4\ldots d_n\ldots$$

where the choices of integers $0 \le d_i \le 9$ can be made in any way? Repeating decimals can always be converted into fractions and so the infinite process can be avoided. But if the pattern does not repeat, a different interpretation must be made.

The most obvious interpretation of this number x is to declare that it is the limit of the sequence

$$\lim_{n\to\infty} 0.d_1d_2d_3d_4\ldots d_n.$$

But how do we know that the limit exists? Our theory provides an immediate answer. Since this sequence is nondecreasing and every term is smaller than 1, by the monotone convergence theorem the sequence converges. This is true no matter what the choices of the decimal digits are. ◀

Example 2.36: (Expansion of e^x) Let $x > 0$ and consider the two closely related sequences

$$s_n = 1+x+\frac{x^2}{2!}+\frac{x^3}{3!}+\cdots+\frac{x^n}{n!}$$

and

$$t_n = \left(1 + \frac{x}{n}\right)^n.$$

The relation between the two sequences becomes more apparent once the binomial theorem is used to expand the latter.

In more advanced mathematics it is shown that both sequences converge to e^x. Let us be content to prove that

$$\lim_{n\to\infty} s_n = \lim_{n\to\infty} t_n.$$

The sequence $\{s_n\}$ is clearly increasing since each new term is the preceding term with a positive number added to it. To show convergence then we need only show that the sequence is bounded. This takes some arithmetic, but not too much.

Choose an integer N larger than $2x$. Note then that

$$\frac{x^{N+1}}{(N+1)!} < \frac{1}{2}\left(\frac{x^N}{N!}\right)$$

that

$$\frac{x^{N+2}}{(N+2)!} < \frac{1}{4}\left(\frac{x^N}{N!}\right)$$

and that

$$\frac{x^{N+3}}{(N+3)!} < \frac{1}{8}\left(\frac{x^N}{N!}\right).$$

Thus

$$\begin{aligned} s_n &\le \left[1 + x + \frac{x^2}{2!} + \cdots + \frac{x^{N-1}}{(N-1)!}\right] + \frac{x^N}{N!}\left(1 + \frac{1}{2} + \frac{1}{4}\cdots\right) \\ &\le \left[1 + x + \frac{x^2}{2!} + \cdots + \frac{x^{N-1}}{(N-1)!}\right] + 2\frac{x^N}{N!}. \end{aligned}$$

Here we have used the limit for the sum of a geometric progression from Example 2.34 to make an upper estimate on how large this sum can get. Note that the N is fixed and so the number on the right-hand side of this inequality is just a number, and it is larger than every number in the sequence $\{s_n\}$.

It follows now from the monotone convergence theorem that $\{s_n\}$ converges. To handle $\{t_n\}$, first apply the binomial theorem to obtain

$$t_n = 1 + x + \frac{1 - 1/n}{2!}x^2 + \frac{(1-1/n)(1-2/n)}{3!}x^3 + \cdots \le s_n.$$

From this we see that $\{t_n\}$ is increasing and that it is smaller than the convergent sequence $\{s_n\}$. It follows, again from the monotone convergence theorem, that $\{t_n\}$ converges. Moreover,

$$\lim_{n\to\infty} t_n \le \lim_{n\to\infty} s_n.$$

If we can obtain the opposite inequality we will have proved our assertion. Let m be a fixed number and let $n > m$. Then, from the preceding expansion, we note that

$$t_n > 1 + x + \frac{1 - 1/n}{2!}x^2 + \frac{(1-1/n)(1-2/n)}{3!}x^3$$

$$+\cdots+\frac{(1-1/n)(1-2/n)+\cdots+(1-[m-1]/n)}{m!}x^m.$$

We can hold m fixed and allow $n\to\infty$ in this inequality and obtain that

$$\lim_{n\to\infty} t_n \geq s_m$$

for each m. From this it now follows that

$$\lim_{n\to\infty} t_n \geq \lim_{n\to\infty} s_n$$

and we have completed our task. ◀

Exercises

2.10.1 Since we know that

$$1+x+x^2+x^3+\cdots+x^n\to\frac{1}{1-x}$$

this suggests the formula

$$1+2+4+8+16+\cdots=\frac{1}{1-2}=-1.$$

Do you have any criticisms?

SEE NOTE 23

2.10.2 Let α and β be positive numbers. Discuss the convergence behavior of the sequence

$$\frac{\alpha^{\beta n}}{\beta^{\alpha n}}.$$

2.10.3 Define

$$e=\lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n.$$

Show that $2<e<3$.

2.10.4 Show that

$$\lim_{n\to\infty}\left(1+\frac{1}{2n}\right)^n=\sqrt{e}.$$

2.10.5 Check the simple identity

$$\left(1+\frac{2}{n}\right)=\left(1+\frac{1}{n+1}\right)\left(1+\frac{1}{n}\right)$$

and use it to show that

$$\lim_{n\to\infty}\left(1+\frac{2}{n}\right)^n=e^2.$$

2.11 Subsequences

The sequence

$$1,-1,2,-2,3,-3,4,-4,5,-5,\ldots$$

appears to contain within itself the two sequences

$$1,2,3,4,5,\ldots$$

and

$$-1,-2,-3,-4,-5,\ldots.$$

In order to have a language to express this we introduce the term *subsequence*. We would say that the latter two sequences are subsequences of the first sequence. Often a sequence is best studied by looking at some of its subsequences. But what is a proper definition of this term? We need a formal mathematical way of expressing the vague idea that a subsequence is obtained by crossing out some of the terms of the original sequence.

Definition 2.37: (Subsequences) Let

$$s_1, s_2, s_3, s_4, \ldots$$

be any sequence. Then by a *subsequence* of this sequence we mean any sequence

$$s_{n_1}, s_{n_2}, s_{n_3}, s_{n_4}, \ldots$$

where

$$n_1 < n_2 < n_3 < \ldots$$

is an increasing sequence of natural numbers.

Example 2.38: We can consider

$$1,2,3,4,5,\ldots$$

to be a subsequence of sequence

$$1,-1,2,-2,3,-3,4,-4,5,-5,\ldots$$

because it contains just the first, third, fifth, etc. terms of the original sequence. Here $n_1 = 1, n_2 = 3, n_3 = 5, \ldots$. ◀

In many applications of sequences it is the subsequences that need to be studied. For example, what can we say about the existence of monotonic subsequences, or bounded subsequences, or divergent subsequences, or convergent subsequences? The answers to these questions have important uses.

Existence of Monotonic Subsequences Our first question is easy to answer for any specific sequence, but harder to settle in general. Given a sequence can we always select a subsequence that is monotonic, either monotonic nondecreasing or monotonic nonincreasing?

Theorem 2.39: *Every sequence contains a monotonic subsequence.*

Proof. We construct first a nonincreasing subsequence if possible. We call the mth element x_m of the sequence $\{x_n\}$ a turn-back point if all later elements are less than or equal to it, in symbols if $x_m \geq x_n$ for all $n > m$. If there is an infinite subsequence of turn-back points $x_{m_1}, x_{m_2}, x_{m_3}, x_{m_4}, \ldots$ then we have found our nonincreasing subsequence since

$$x_{m_1} \geq x_{m_2} \geq x_{m_3} \geq x_{m_4} \geq \ldots.$$

This would not be possible if there are only finitely many turn-back points. Let us suppose that x_M is the last turn-back point so that any element x_n for $n > M$ is not a turn-back point. Since it is not there must be an element further on in the sequence greater than it, in symbols $x_m > x_n$ for some $m > n$. Thus we can choose $x_{m_1} > x_{M+1}$ with $m_1 > M+1$, then $x_{m_2} > x_{m_1}$ with $m_2 > m_1$, and then $x_{m_3} > x_{m_2}$ with $m_3 > m_2$, and so on to obtain an increasing subsequence

$$x_{M+1} < x_{m_1} < x_{m_2} < x_{m_3} < x_{m_4} < \dots$$

as required. ■

Existence of Convergent Subsequences Having answered this question about the existence of monotonic subsequences, we can also now answer the question about the existence of convergent subsequences. This might, at first sight, seem just a curiosity, but it will give us later one of our most important tools in analysis.

The theorem is traditionally attributed to two major nineteenth-century mathematicians, Karl Theodor Wilhelm Weierstrass (1815-1897) and Bernhard Bolzano (1781–1848). These two mathematicians, the first German and the second Czech, rank with Cauchy among the founders of our subject.

Theorem 2.40 (Bolzano-Weierstrass) *Every bounded sequence contains a convergent subsequence.*

Proof. By Theorem 2.39 every sequence contains a monotonic subsequence. Here that subsequence would be both monotonic and bounded, and hence convergent. ■

Other (less important) questions of this type appear in the exercises.

Exercises

2.11.1 Show that, according to our definition, every sequence is a subsequence of itself. How would the definition have to be reworded to avoid this if, for some reason, this possibility were to have been avoided?

2.11.2 Show that every subsequence of a subsequence of a sequence $\{x_n\}$ is itself a subsequence of $\{x_n\}$.

2.11.3 If $\{s_{n_k}\}$ is a subsequence of $\{s_n\}$ and $\{t_{m_k}\}$ is a subsequence of $\{t_n\}$ then is it true that $\{s_{n_k} + t_{m_k}\}$ is a subsequence of $\{s_n + t_n\}$?

2.11.4 If $\{s_{n_k}\}$ is a subsequence of $\{s_n\}$ is $\{(s_{n_k})^2\}$ a subsequence of $\{(s_n)^2\}$?

2.11.5 Describe all sequences that have only finitely many different subsequences.

2.11.6 Establish which of the following statements are true.

(a) A sequence is convergent if and only if all of its subsequences are convergent.

(b) A sequence is bounded if and only if all of its subsequences are bounded.

(c) A sequence is monotonic if and only if all of its subsequences are monotonic.

(d) A sequence is divergent if and only if all of its subsequences are divergent.

2.11.7 Establish which of the following statements are true for an arbitrary sequence $\{s_n\}$.

(a) If all monotone subsequences of a sequence $\{s_n\}$ are convergent, then $\{s_n\}$ is bounded.

(b) If all monotone subsequences of a sequence $\{s_n\}$ are convergent, then $\{s_n\}$ is convergent.

(c) If all convergent subsequences of a sequence $\{s_n\}$ converge to 0, then $\{s_n\}$ converges to 0.

(d) If all convergent subsequences of a sequence $\{s_n\}$ converge to 0 and $\{s_n\}$ is bounded, then $\{s_n\}$ converges to 0.

2.11.8 Where possible find subsequences that are monotonic and subsequences that are convergent for the following sequences

(a) $\{(-1)^n n\}$

(b) $\{\sin(n\pi/8)\}$

(c) $\{n\sin(n\pi/8)\}$

(d) $\left\{\frac{n+1}{n}\sin(n\pi/8)\right\}$

(e) $\{1+(-1)^n\}$

(f) $\{r_n\}$ consists of all rational numbers in the interval $(0,1)$ arranged in some order.

2.11.9 Describe all subsequences of the sequence

$$1,0,1,0,1,0,1,0,1,0,1,0,\ldots.$$

Describe all convergent subsequences. Describe all monotonic subsequences.

2.11.10 If $\{s_{n_k}\}$ is a subsequence of $\{s_n\}$ show that $n_k \geq k$ for all $k = 1,2,3,\ldots$.

2.11.11 Give an example of a sequence that contains subsequences converging to every natural number (and no other numbers).

2.11.12 Give an example of a sequence that contains subsequences converging to every number in $[0,1]$ (and no other numbers).

2.11.13 Show that there cannot exist a sequence that contains subsequences converging to every number in $(0,1)$ and no other numbers.

SEE NOTE 24

2.11.14 Show that if $\{s_n\}$ has no convergent subsequences, then $|s_n| \to \infty$ as $n \to \infty$.

2.11.15 If a sequence $\{x_n\}$ has the property that

$$\lim_{n\to\infty} x_{2n} = \lim_{n\to\infty} x_{2n+1} = L$$

show that the sequence $\{x_n\}$ converges to L.

2.11.16 If a sequence $\{x_n\}$ has the property that

$$\lim_{n\to\infty} x_{2n} = \lim_{n\to\infty} x_{2n+1} = \infty$$

show that the sequence $\{x_n\}$ diverges to ∞.

2.11.17 Let α and β be positive real numbers and define a sequence by setting $s_1 = \alpha$, $s_2 = \beta$ and $s_{n+2} = \frac{1}{2}(s_n + s_{n+1})$ for all $n = 1,2,3,\ldots$. Show that the subsequences $\{s_{2n}\}$ and $\{s_{2n-1}\}$ are monotonic and convergent. Does the sequence $\{s_n\}$ converge? To what?

2.11.18 Without appealing to any of the theory of this section prove that every unbounded sequence has a strictly monotonic subsequence (i.e., either increasing or decreasing).

2.11.19 Show that if a sequence $\{x_n\}$ converges to a finite limit or diverges to $\pm\infty$ then every subsequence has precisely the same behavior.

2.11.20 Suppose a sequence $\{x_n\}$ has the property that every subsequence has a further subsequence convergent to L. Show that $\{x_n\}$ converges to L.

2.11.21 Let $\{x_n\}$ be a bounded sequence and let $x = \sup\{x_n : n \in \mathbb{N}\}$. Suppose that, moreover, $x_n < x$ for all n. Prove that there is a subsequence convergent to x.

2.11.22 Let $\{x_n\}$ be a bounded sequence, let

$$y = \inf\{x_n : n \in \mathbb{N}\} \text{ and } x = \sup\{x_n : n \in \mathbb{N}\}.$$

Suppose that, moreover, $y < x_n < x$ for all n. Prove that there is a pair of convergent subsequences $\{x_{n_k}\}$ and $\{x_{m_k}\}$ so that

$$\lim_{k\to\infty} \left|x_{n_k} - x_{m_k}\right| = x - y.$$

2.11.23 Does every divergent sequence contain a divergent monotonic subsequence?

2.11.24 Does every divergent sequence contain a divergent bounded subsequence?

2.11.25 Construct a proof of the Bolzano-Weierstrass theorem for bounded sequences using the nested interval property and not appealing to the existence of monotonic subsequences.

2.11.26 Construct a direct proof of the assertion that every convergent sequence has a convergent, monotonic subsequence (i.e., without appealing to Theorem 2.39).

2.11.27 Let $\{x_n\}$ be a bounded sequence that we do not know converges. Suppose that it has the property that every one of its convergent subsequences converges to the same number L. What can you conclude?

2.11.28 Let $\{x_n\}$ be a bounded sequence that diverges. Show that there is a pair of convergent subsequences $\{x_{n_k}\}$ and $\{x_{m_k}\}$ so that

$$\lim_{k\to\infty} \left|x_{n_k} - x_{m_k}\right| > 0.$$

2.11.29 Let $\{x_n\}$ be a sequence. A number z with the property that for all $\varepsilon > 0$ there are infinitely many terms of the sequence in the interval $(z-\varepsilon, z+\varepsilon)$ is said to be a *cluster point* of the sequence. Show that z is a cluster point of a sequence if and only if there is a subsequence $\{x_{n_k}\}$ converging to z.

2.12 Cauchy Convergence Criterion

What property of a sequence characterizes convergence? As a "characterization" we would like some necessary and sufficient condition for a sequence to converge. We could simply write the definition and consider that that is a characterization. Thus the following technical statement would, indeed, be a characterization of the convergence of a sequence $\{s_n\}$.

A sequence $\{s_n\}$ is convergent if and only if $\exists L$ so that $\forall \varepsilon > 0$ $\exists N$ with the property that

$$|s_n - L| < \varepsilon$$

whenever $n \geq N$.

In mathematics when we ask for a characterization of a property we can expect to find many answers, some more useful than others. The limitation of this particular characterization is that it requires us to find the number L which is the limit of the sequence in advance. Compare this with a characterization of convergence of a monotonic sequence $\{s_n\}$.

A monotonic sequence $\{s_n\}$ is convergent if and only if it is bounded.

This is a wonderful and most useful characterization. But it applies only to monotonic sequences.

A correct and useful characterization, applicable to all sequences, was found by Cauchy. This is the content of the next theorem. Note that it has the advantage that it describes a convergent sequence with no reference whatsoever to the actual value of the limit. Loosely it asserts that a sequence converges if and only if the terms of the sequence are eventually arbitrarily close together.

Theorem 2.41 (Cauchy Criterion) *A sequence $\{s_n\}$ is convergent if and only if for each $\varepsilon > 0$ there exists an integer N with the property that*

$$|s_n - s_m| < \varepsilon$$

whenever $n \geq N$ and $m \geq N$.

Proof. This property of the theorem is so important that it deserves some terminology. A sequence is said to be a *Cauchy sequence* if it satisfies this property. Thus the theorem states that a sequence is convergent if and only if it is a Cauchy sequence. The terminology is most significant in more advanced situations where being a Cauchy sequence is not necessarily equivalent with being convergent.

Our proof is a bit lengthy and will require an application of the Bolzano-Weierstrass theorem.

The proof in one direction, however, is easy. Suppose that $\{s_n\}$ is convergent to a number L. Let $\varepsilon > 0$. Then there must be an integer N so that

$$|s_k - L| < \frac{\varepsilon}{2}$$

whenever $k \geq N$. Thus if both m and n are larger than N,

$$|s_n - s_m| \leq |s_n - L| + |L - s_m| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$$

which shows that $\{s_n\}$ is a Cauchy sequence.

Now let us prove the opposite (and more difficult) direction.

For the first step we show that every Cauchy sequence is bounded. Since the proof of this can be obtained by copying and modifying the proof of Theorem 2.11, we have

left this as an exercise. (It is not really interesting that Cauchy sequences are bounded since after the proof is completed we know that all Cauchy sequences are convergent and so must, indeed, be bounded.)

For the second step we apply the Bolzano-Weierstrass theorem to the bounded sequence $\{s_n\}$ to obtain a convergent subsequence $\{s_{n_k}\}$.

The final step is a feature of Cauchy sequences. Once we know that $s_{n_k} \to L$ and that $\{s_n\}$ is Cauchy, we can show that $s_n \to L$ also. Let $\varepsilon > 0$ and choose N so that

$$|s_n - s_m| < \varepsilon/2$$

for all $m, n \geq N$. Choose K so that

$$|s_{n_k} - L| < \varepsilon/2$$

for all $k \geq K$. Suppose that $n \geq N$. Set m equal to any value of n_k that is larger than N and so that $k \geq K$. For this value $s_m = s_{n_k}$

$$|s_n - L| \leq |s_n - s_{n_k}| + |s_{n_k} - L| < \varepsilon/2 + \varepsilon/2 = \varepsilon.$$

By definition, $\{s_n\}$ converges to L and so the proof is complete. ■

Example 2.42: The Cauchy criterion is most useful in theoretical developments rather than applied to concrete examples. Even so, occasionally it is the fastest route to a proof of convergence. For example, consider the sequence $\{x_n\}$ defined by setting $x_1 = 1, x_2 = 2$ and then, recursively,

$$x_n = \frac{x_{n-1} + x_{n-2}}{2}.$$

Each term after the second is the average of the preceding two terms. The distance between x_1 and x_2 is 1, that between x_2 and x_3 is $1/2$, between x_3 and x_4 is $1/4$, and so on. We see then that after the N stage all the distances are smaller than 2^{-N+1}, that is, that for all $n \geq N$ and $m \geq N$

$$|x_n - x_m| \leq \frac{1}{2^{N-1}}.$$

This is exactly the Cauchy criterion and so this sequence converges. Note that the Cauchy criterion offers no information on what the sequence is converging to. You must come up with another method to find out. ◀

Exercises

2.12.1 Show directly that the sequence $s_n = 1/n$ is a Cauchy sequence.

2.12.2 Show directly that any multiple of a Cauchy sequence is again a Cauchy sequence.

2.12.3 Show directly that the sum of two Cauchy sequences is again a Cauchy sequence.

2.12.4 Show directly that any Cauchy sequence is bounded.

2.12.5 The following criterion is weaker than the Cauchy criterion. Show that it is not equivalent:

For all $\varepsilon > 0$ there exists an integer N with the property that

$$|s_{n+1} - s_n| < \varepsilon$$

whenever $n \geq N$.

SEE NOTE 25

2.12.6 A careless student believes that the following statement is the Cauchy criterion.

For all $\varepsilon > 0$ and all positive integers p there exists an integer N with the property that

$$|s_{n+p} - s_n| < \varepsilon$$

whenever $n \geq N$.

Is this statement weaker, stronger, or equivalent to the Cauchy criterion?

2.12.7 Show directly that if $\{s_n\}$ is a Cauchy sequence then so too is $\{|s_n|\}$. From this conclude that $\{|s_n|\}$ converges whenever $\{s_n\}$ converges.

2.12.8 Show that every subsequence of a Cauchy sequence is Cauchy. (Do not use the fact that every Cauchy sequence is convergent.)

2.12.9 Show that every bounded monotonic sequence is Cauchy. (Do not use the monotone convergence theorem.)

2.12.10 Show that the sequence in Example 2.42 converges to $5/3$.

SEE NOTE 26

2.13 Upper and Lower Limits

✂ *Advanced section. May be omitted.*

If $\lim_{n\to\infty} x_n = L$ then, according to our definition, numbers α and β on either side of L, that is, $\alpha < L < \beta$, have the property that

$$\alpha < x_n \text{ and } x_n < \beta$$

for all sufficiently large n. In many applications only *half* of this information is used.

Example 2.43: Here is an example showing how half a limit is as good as a whole limit. Let $\{x_n\}$ be a sequence of positive numbers with the property that

$$\lim_{n\to\infty} \sqrt[n]{x_n} = L < 1.$$

Then we can prove that $x_n \to 0$. To see this pick numbers α and β so that

$$\alpha < L < \beta < 1.$$

There must be an integer N so that

$$\alpha < \sqrt[n]{x_n} < \beta < 1$$

for all $n \geq N$. Forget half of this and focus on

$$\sqrt[n]{x_n} < \beta < 1.$$

Then we have

$$x_n < \beta^n$$

for all $n \geq N$ and it is clear now why $x_n \to 0$. ◀

This example suggests that the definition of limit might be weakened to handle situations where less is needed. This way we have a tool to discuss the limiting behavior of sequences that may not necessarily converge. Even if the sequence does converge this often offers a tool that can be used without first finding a proof of convergence.

We break the definition of sequence limit into two half-limits as follows.

Definition 2.44: (Lim Sup) A *limit superior* of a sequence $\{x_n\}$, denoted as

$$\limsup_{n\to\infty} x_n,$$

is defined to be the infimum of all numbers β with the following property:

There is an integer N so that $x_n < \beta$ for all $n \geq N$.

Definition 2.45: (Lim Inf) A *limit inferior* of a sequence $\{x_n\}$, denoted as

$$\liminf_{n\to\infty} x_n,$$

is defined to be the supremum of all numbers α with the following property:

There is an integer N so that $\alpha < x_n$ for all $n \geq N$.

Note. In interpreting this definition note that, by our usual rules on infs and sups, the values $-\infty$ and ∞ are allowed. If there are *no* numbers β with the property of the definition, then the sequence is simply unbounded above. The infimum of the empty set is taken as ∞ and so

$$\limsup_{n\to\infty} x_n = \infty \Leftrightarrow \text{the sequence } \{x_n\} \text{ has no upper bound.}$$

On the other hand, if *every* number β has the property of the definition this means exactly that our sequence must be diverging to $-\infty$. The infimum of the set of *all* real numbers is taken as $-\infty$ and so

$$\limsup_{n\to\infty} x_n = -\infty \Leftrightarrow \text{the sequence } \{x_n\} \to -\infty.$$

The same holds in the other direction. A sequence that is unbounded below can be described by saying $\liminf_{n\to\infty} x_n = -\infty$. A sequence that diverges to ∞ can be described by saying $\liminf_{n\to\infty} x_n = \infty$.

We refer to these concepts as "upper limits" and "lower limits" or "extreme limits." They extend our theory describing the limiting behavior of sequences to allow precise descriptions of divergent sequences. Obviously, we should establish very quickly that the upper limit is indeed greater than or equal to the lower limit since our language suggests this.

Theorem 2.46: *Let $\{x_n\}$ be a sequence of real numbers. Then*

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n.$$

Proof. If $\limsup_{n\to\infty} x_n = \infty$ or if $\liminf_{n\to\infty} x_n = -\infty$ we have nothing to prove. If not then take any number β larger than $\limsup_{n\to\infty} x_n$ and any number α smaller than $\liminf_{n\to\infty} x_n$. By definition then there is an integer N so that $x_n < \beta$ for all $n \geq N$ and an integer M so that $\alpha < x_n$ for all $n \geq M$. It must be true that $\alpha < \beta$. But β is *any* number larger than $\limsup_{n\to\infty} x_n$. Hence

$$\alpha \leq \limsup_{n\to\infty} x_n.$$

Similarly, α is *any* number smaller than $\liminf_{n\to\infty} x_n$. Hence

$$\liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n$$

as required. ■

How shall we use the limit superior of a sequence $\{x_n\}$? If

$$\limsup_{n\to\infty} x_n = L$$

then every number $\beta > L$ has the property that $x_n < \beta$ for all n large enough. This is because L is the infimum of such numbers β. On the other hand, any number $b < L$ cannot have this property so $x_n \geq b$ for infinitely many indices n. Thus numbers slightly larger than L must be upper bounds for the sequence eventually. Numbers slightly less than L are not upper bounds eventually. To express this a little more precisely, the number L is the limit superior of a sequence $\{x_n\}$ exactly when the following holds:

> For every $\varepsilon > 0$ there is an integer N so that $x_n < L + \varepsilon$ for all $n \geq N$ and $x_n > L - \varepsilon$ for infinitely many $n \geq N$.

The next theorem gives another characterization which is sometimes easier to apply. This version also better explains why we describe this notion as a "lim sup" and "lim inf."

Theorem 2.47: *Let $\{x_n\}$ be a sequence of real numbers. Then*

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\}$$

and

$$\liminf_{n\to\infty} x_n = \lim_{n\to\infty} \inf\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\}.$$

Proof. Let us prove just the statement for lim sups as the lim inf statement can be proved similarly.

Write

$$y_n = \sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\}.$$

Then $x_n \leq y_n$ for all n and so, using the inequality promised in Exercise 2.13.5,

$$\limsup_{n\to\infty} x_n \leq \limsup_{n\to\infty} y_n.$$

But $\{y_n\}$ is a nonincreasing sequence and so

$$\limsup_{n\to\infty} y_n = \lim_{n\to\infty} y_n.$$

From this it follows that

$$\limsup_{n\to\infty} x_n \leq \lim_{n\to\infty} \sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\}.$$

Let us now show the reverse inequality. If $\limsup_{n\to\infty} x_n = \infty$ then the sequence is unbounded above. Thus for all n

$$\sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\} = \infty$$

and so, in this case,

$$\limsup_{n\to\infty} x_n = \lim_{n\to\infty} \sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\}$$

must certainly be true.

If

$$\limsup_{n\to\infty} x_n < \infty$$

then take any number β larger than $\limsup_{n\to\infty} x_n$. By definition then there is an integer N so that $x_n < \beta$ for all $n \geq N$. It follows that

$$\lim_{n\to\infty} \sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\} \leq \beta.$$

But β is *any* number larger than $\limsup_{n\to\infty} x_n$. Hence

$$\lim_{n\to\infty} \sup\{x_n, x_{n+1}, x_{n+2}, x_{n+3}, \dots\} \leq \limsup_{n\to\infty} x_n.$$

We have proved both inequalities, the equality follows, and the theorem is proved. ■

The connection between limits and extreme limits is close. If a limit exists then the upper and lower limits must be the same.

Theorem 2.48: *Let $\{x_n\}$ be a sequence of real numbers. Then $\{x_n\}$ is convergent if and only if $\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n$ and these are finite. In this case*

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = \lim_{n\to\infty} x_n.$$

Proof. Let $\varepsilon > 0$. If $\limsup_{n\to\infty} x_n = L$ then there is an integer N_1 so that $x_n < L + \varepsilon$ for all $n \geq N_1$. If it is also true that $\liminf_{n\to\infty} x_n = L$ then there is an integer N_2 so that $x_n > L - \varepsilon$ for all $n \geq N_2$. Putting these together we have

$$L - \varepsilon < x_n < L + \varepsilon$$

for all

$$n \geq N = \max\{N_1, N_2\}.$$

By definition then $\lim_{n\to\infty} x_n = L$.

Conversely, if $\lim_{n\to\infty} x_n = L$ then for some N,

$$L - \varepsilon < x_n < L + \varepsilon$$

for all $n \geq N$. Thus

$$L - \varepsilon \leq \liminf_{n\to\infty} x_n \leq \limsup_{n\to\infty} x_n \leq L + \varepsilon.$$

Since ε is an arbitrary positive number we must have

$$L = \liminf_{n\to\infty} x_n = \limsup_{n\to\infty} x_n$$

as required. ■

In the exercises you will be asked to compute several lim sups and lim infs. This is just for familiarity with the concepts. Computations are not so important. What is important is the use of these ideas in theoretical developments. More critical is how these limit operations relate to arithmetic or order properties. The limit of a sum is the sum of the two limits. Is this true for lim sups and lim infs? (See Exercise 2.13.9.) Do not skip these exercises.

Exercises

2.13.1 Complete Example 2.43 by showing that if $\{x_n\}$ is a sequence of positive numbers with the property that $\limsup_{n\to\infty} \sqrt[n]{x_n} < 1$ then $x_n \to 0$. Show that if

$$\liminf_{n\to\infty} \sqrt[n]{x_n} > 1$$

then $x_n \to \infty$. What can you conclude if $\limsup_{n\to\infty} \sqrt[n]{x_n} > 1$? What can you conclude if $\liminf_{n\to\infty} \sqrt[n]{x_n} < 1$?

2.13.2 Compute lim sups and lim infs for the following sequences

(a) $\{(-1)^n n\}$

(b) $\{\sin(n\pi/8)\}$

(c) $\{n\sin(n\pi/8)\}$

(d) $\{[(n+1)\sin(n\pi/8)]/n\}$

(e) $\{1+(-1)^n\}$

(f) $\{r_n\}$ consists of all rational numbers in the interval $(0,1)$ arranged in some order.

2.13.3 Give examples of sequences of rational numbers $\{a_n\}$ with

(a) upper limit $\sqrt{2}$ and lower limit $-\sqrt{2}$,

(b) upper limit $+\infty$ and lower limit $\sqrt{2}$,

(c) upper limit π and lower limit e.

2.13.4 Show that $\limsup_{n\to\infty}(-x_n) = -(\liminf_{n\to\infty} x_n)$.

2.13.5 If two sequences $\{a_n\}$ and $\{b_n\}$ satisfy the inequality $a_n \le b_n$ for all sufficiently large n, show that

$$\limsup_{n\to\infty} a_n \le \limsup_{n\to\infty} b_n \quad \text{and} \quad \liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} b_n.$$

2.13.6 Show that $\lim_{n\to\infty} x_n = \infty$ if and only if

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = \infty.$$

2.13.7 Show that if $\limsup_{n\to\infty} a_n = L$ for a finite real number L and $\varepsilon > 0$, then $a_n > L+\varepsilon$ for only finitely many n and $a_n > L-\varepsilon$ for infinitely many n.

2.13.8 Show that for any monotonic sequence $\{x_n\}$

$$\limsup_{n\to\infty} x_n = \liminf_{n\to\infty} x_n = \lim_{n\to\infty} x_n$$

(including the possibility of infinite limits).

2.13.9 Show that for any bounded sequences $\{a_n\}$ and $\{b_n\}$

$$\limsup_{n\to\infty}(a_n+b_n) \le \limsup_{n\to\infty} a_n + \limsup_{n\to\infty} b_n.$$

Give an example to show that the equality need not occur.

2.13.10 What is the correct version for the lim inf of Exercise 2.13.9?

2.13.11 Show that for any bounded sequences $\{a_n\}$ and $\{b_n\}$ of positive numbers

$$\limsup_{n\to\infty}(a_n b_n) \le (\limsup_{n\to\infty} a_n)(\limsup_{n\to\infty} b_n).$$

Give an example to show that the equality need not occur.

2.13.12 Correct the careless student proof in Exercise 2.8.3 for the squeeze theorem by replacing lim with limsup and liminf in the argument.

2.13.13 What relation, if any, can you state for the lim sups and lim infs of a sequence $\{a_n\}$ and one of its subsequences $\{a_{n_k}\}$?

2.13.14 If a sequence $\{a_n\}$ has no convergent subsequences, what can you state about the lim sups and lim infs of the sequence?

2.13.15 Let S denote the set of all real numbers t with the property that some subsequence of a given sequence $\{a_n\}$ converges to t. What is the relation between the set S and the lim sups and lim infs of the sequence $\{a_n\}$?

SEE NOTE 27

2.13.16 Prove the following assertion about the upper and lower limits for any sequence $\{a_n\}$ of positive real numbers:

$$\liminf_{n\to\infty} \frac{a_{n+1}}{a_n} \le \liminf_{n\to\infty} \sqrt[n]{a_n} \le \limsup_{n\to\infty} \sqrt[n]{a_n} \le \limsup_{n\to\infty} \frac{a_{n+1}}{a_n}.$$

Give an example to show that each of these inequalities may be strict.

2.13.17 For any sequence $\{a_n\}$ write $s_n = (a_1 + a_2 + \cdots + a_n)/n$. Show that

$$\liminf_{n\to\infty} a_n \le \liminf_{n\to\infty} s_n \le \limsup_{n\to\infty} s_n \le \limsup_{n\to\infty} a_n.$$

Give an example to show that each of these inequalities may be strict.

2.14 Challenging Problems for Chapter 2

2.14.1 Let α and β be positive numbers. Show that

$$\lim_{n\to\infty} \sqrt[n]{\alpha^n + \beta^n} = \max\{\alpha, \beta\}.$$

2.14.2 For any convergent sequence $\{a_n\}$ write $s_n = (a_1 + a_2 + \cdots + a_n)/n$, the sequence of averages. Show that

$$\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n.$$

Give an example to show that $\{s_n\}$ could converge even if $\{a_n\}$ diverges.

2.14.3 Let $a_1 = 1$ and define a sequence recursively by

$$a_{n+1} = \sqrt{a_1 + a_2 + \cdots + a_n}.$$

Show that $\lim_{n\to\infty} \frac{a_n}{n} = 1/2$.

2.14.4 Let $x_1 = \theta$ and define a sequence recursively by

$$x_{n+1} = \frac{x_n}{1 + x_n/2}.$$

For what values of θ is it true that $x_n \to 0$?

2.14.5 Let $\{a_n\}$ be a sequence of numbers in the interval $(0,1)$ with the property that

$$a_n < \frac{a_{n-1} + a_{n+1}}{2}$$

for all $n = 2, 3, 4, \ldots$. Show that this sequence is convergent.

2.14.6 For any convergent sequence $\{a_n\}$ write

$$s_n = \sqrt[n]{(a_1 a_2 \ldots a_n)},$$

the sequence of geometric averages. Show that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} s_n$. Give an example to show that $\{s_n\}$ could converge even if $\{a_n\}$ diverges.

2.14.7 If

$$\lim_{n\to\infty} \frac{s_n - \alpha}{s_n + \alpha} = 0$$

what can you conclude about the sequence $\{s_n\}$?

2.14.8 A function f is defined by

$$f(x) = \lim_{n\to\infty} \left(\frac{1 - x^2}{1 + x^2} \right)^n$$

at every value x for which this limit exists. What is the domain of the function?

2.14.9 A function f is defined by

$$f(x) = \lim_{n\to\infty} \frac{1}{x^n + x^{-n}}$$

at every value x for which this limit exists. What is the domain of the function?

2.14.10 Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a positive function with a derivative f' that is everywhere continuous and negative. Apply Newton's method to obtain a sequence

$$x_1 = \theta, \;\; x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Show that $x_n \to \infty$ for any starting value θ. [This problem assumes some calculus background.]

2.14.11 Let $f(x) = x^3 - 3x + 3$. Apply Newton's method to obtain a sequence

$$x_1 = \theta, \;\; x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}.$$

Show that for any positive integer p there is a starting value θ such that the sequence $\{x_n\}$ is periodic with period p.

SEE NOTE 28

2.14.12 Determine all subsequential limit points of the sequence $x_n = \cos n$.

SEE NOTE 29

2.14.13 A sequence $\{s_n\}$ is said to be *contractive* if there is a positive number $0 < r < 1$ so that

$$|s_{n+1} - s_n| \leq r|s_n - s_{n-1}|$$

for all $n = 2, 3, 4, \ldots$.

(a) Show that the sequence defined by $s_1 = 1$ and $s_n = (4 + s_{n-1})^{-1}$ for $n = 2, 3, \ldots$ is contractive.

(b) Show that every contractive sequence is Cauchy.

(c) Show that a sequence can satisfy the condition

$$|s_{n+1} - s_n| < |s_n - s_{n-1}|$$

for all $n = 2, 3, 4, \ldots$ and not be contractive, nor even convergent.

(d) Is every convergent sequence contractive?

SEE NOTE 30

2.14.14 The sequence defined recursively by

$$f_1 = 1,\ f_2 = 1 \quad f_{n+2} = f_n + f_{n+1}$$

is called the *Fibonacci sequence*. Let

$$r_n = f_{n+1}/f_n$$

be the sequence of ratios of successive terms of the Fibonacci sequence.

(a) Show that $r_1 < r_3 < r_5 < \cdots < r_6 < r_4 < r_2$.

(b) Show that $r_{2n} - r_{2n-1} \to 0$.

(c) Deduce that the sequence $\{r_n\}$ converges. Can you find a way to determine that limit? (This is related to the roots of the equation $x^2 - x - 1 = 0$.)

2.14.15 A sequence of real numbers $\{x_n\}$ has the property that

$$(2 - x_n)x_{n+1} = 1.$$

Show that $\lim_{n\to\infty} x_n = 1$.

SEE NOTE 31

2.14.16 Let $\{a_n\}$ be an arbitrary sequence of positive real numbers. Show that

$$\limsup_{n\to\infty} \left(\frac{a_1 + a_{n+1}}{a_n} \right)^n \geq e.$$

SEE NOTE 32

2.14.17 Suppose that the sequence whose nth term is

$$s_n + 2s_{n+1}$$

is convergent. Show that $\{s_n\}$ is also convergent.

SEE NOTE 33

2.14.18 Show that the sequence

$$\sqrt{7}, \sqrt{7 - \sqrt{7}}, \sqrt{7 - \sqrt{7 + \sqrt{7}}}, \sqrt{7 - \sqrt{7 + \sqrt{7 - \sqrt{7}}}}, \ldots$$

converges and find its limit.

SEE NOTE 34

2.14.19 Let a_1 and a_2 be positive numbers and suppose that the sequence $\{a_n\}$ is defined recursively by

$$a_{n+2} = \sqrt{a_n} + \sqrt{a_{n+1}}.$$

Show that this sequence converges and find its limit.

SEE NOTE 35

Notes

[9]Exercise 2.2.2. For the next term in the sequence some people might expect a 1. Most mathematicians would expect a 9.

[10]Exercise 2.2.3. Here is a formula that generates the first five terms of the sequence 0, 0, 0, 0, c, ...

$$f(n) = \frac{c(n-1)(n-2)(n-3)(n-4)}{4!}.$$

[11]Exercise 2.2.10. The formula is

$$f_n = \frac{1}{\sqrt{5}}\left\{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n\right\}.$$

It can be verified by induction.

[12]Exercise 2.3.1. Find a function $f : (a,b) \to (0,1)$ one-to-one onto and consider the sequence $f(s_n)$, where $\{s_n\}$ is a sequence that is claimed to have all of (a,b) as its range.

[13]Exercise 2.3.4. We can consider that the elements of each of the sets S_i can be listed, say,

$$S_1 = \{x_{11}, x_{12}, x_{13}, \dots\}$$

$$S_2 = \{x_{21}, x_{22}, x_{23}, \dots\}$$

and so on. Now try to think of a way of listing all of these items, that is, making one big list that contains them all.

[14]Exercise 2.3.6. We need (i) every number has a decimal expansion; (ii) the decimal expansion is unique except in the case of expansions that terminate in a string of zeros or nines [e.g., $1/2 = 0.5000000\cdots = .49999999\dots$], thus if a and b are numbers such that in the nth decimal place one has a 5 (or a 6) and the other does not then either $a \neq b$, or perhaps one ends in a string of zeros and the other in a string of nines; and (iii) every string of 5's and 6's defines a real number with that decimal expansion.

[15]Exercise 2.3.10. Try to find a way of ranking the algebraic numbers in the same way that the rational numbers were ranked.

[16]Exercise 2.4.6. You will need the identity

$$1+2+3+\cdots+n = n(n+1)/2.$$

[17]Exercise 2.4.7. You will need to find an identity for the sum of the squares similar to the identity $1+2+3+\cdots+n = n(n+1)/2$.

[18]Exercise 2.5.6. To establish a correct converse, reword: If all $x_n > 0$ and $\frac{x_n}{x_n+1} \to 1$, then $x_n \to \infty$. Prove that this is true. The converse of the statement in the exercise is false (e.g., $x_n = 1/n$).

[19]Exercise 2.6.5. Use the same method as used in the proof of Theorem 2.11.

[20]Exercise 2.8.1. Give a counterexample. Perhaps find two sequences so that $s_n < 0 < t_n$ for all n and yet $\lim_{n\to\infty} s_n = \lim_{n\to\infty} t_n = 0$.

[21]Exercise 2.8.9. Take any number r strictly between 1 and that limit. Show that for some N, $s_{n+1} < rs_n$ if $n \geq N$. Deduce that

$$s_{N+2} < r^2 s_N$$

and

$$s_{N+3} < r^3 s_N.$$

Carry on.

[22]Exercise 2.8.10. Take any number r strictly between 1 and that limit. Show that for some N, $s_{n+1} > rs_n$ if $n \geq N$. Deduce that

$$s_{N+2} > r^2 s_N$$

and

$$s_{N+3} > r^3 s_N.$$

Carry on.

[23]Exercise 2.10.1. In terms of our theory of convergence this statement has no meaning since (as you should show) the sequence diverges. Even so, many great mathematicians, including Euler, would have accepted and used this formula. The fact that it is useful suggests that there are ways of interpreting such statements other than as convergence assertions.

[24]Exercise 2.11.13. If a sequence contains subsequences converging to every number in $(0,1)$ show that it also contains a subsequence converging to 0.

[25]Exercise 2.12.5. Consider the sequence

$$s_n = 1 + 1/2 + 1/3 + \cdots + 1/n.$$

[26]Exercise 2.12.10. Compare to

$$1 + 1 - \frac{1}{2} + \frac{1}{4} - \frac{1}{8} + \frac{1}{6} - \ldots$$

which is the sum of a geometric progression.

[27]Exercise 2.13.15. Consider separately the cases where the sequence is bounded or not.

[28]Exercise 2.14.11. A sequence $\{x_n\}$ is periodic with period p if $x_{n+p} = x_n$ for all values of n and no smaller value of p will work. (Note that if $\{x_n\}$ is periodic with period p, then $x_n = x_{n+p} = x_{n+2p} = x_{n+3p} = \ldots.$)

[29]Exercise 2.14.12. Clearly, no number larger than 1 or less than -1 could be such a limit. Show that in fact the interval $[-1,1]$ is the set of all such limit points. If $x \in [-1,1]$ there must be a number y so that $\cos y = x$ (why?). Now consider the set of numbers

$$G = \{n + 2m\pi : n, m \in \mathbb{Z}\}.$$

Using Exercise 1.11.6 or otherwise, show that this is dense. Hence there are pairs of integers n, m so that

$$|y - n + 2m\pi| < \varepsilon.$$

From this deduce that

$$|\cos y - \cos(n + 2m\pi)| < \varepsilon$$

and so $|x - \cos n| < \varepsilon$.

[30]Exercise 2.14.13. For (a) show that

$$|s_{n+1} - s_n| \leq \frac{1}{17}|s_n - s_{n-1}|$$

for all $n = 2, 3, 4, \ldots$. For (b) you will need to use the fact that the sum of geometric progressions is bounded, in fact that

$$1 + r + r^2 + \cdots + r^n < (1-r)^{-1}$$

if $0 < r < 1$. Express for $m > n$,

$$|s_m - s_n| \leq |s_{n+1} - s_n|$$
$$+|s_{n+2} - s_{n+1}| + \cdots + |s_m - s_{m-1}|$$

and then use the contractive hypothesis. Note that

$$|s_4 - s_3| \leq r|s_3 - s_2| \leq r^2|s_2 - s_1|.$$

For (d) you might have to wait for the study of series in order to find an appropriate example of a convergent sequence that is not contractive.

[31]Exercise 2.14.15. This is from the 1947 Putnam Mathematical Competition.

[32]Exercise 2.14.16. This is from the 1949 Putnam Mathematical Competition.

[33]Exercise 2.14.17. This is from the 1950 Putnam Mathematical Competition.

[34]Exercise 2.14.18. This is from the 1953 Putnam Mathematical Competition.

[35]Exercise 2.14.19. Problem posed by A. Emerson in the *Amer. Math. Monthly*, **85** (1978), p. 496.

Chapter 3

INFINITE SUMS

✂ This chapter on infinite sums and series may be skipped over in designing a course or covered later as the need arises. The basic material in Sections 3.4, 3.5, and parts of 3.6 will be needed, but not before the study of series of functions in Chapter 9. All of the enrichment or advanced sections may be omitted and are not needed in the sequel.

3.1 Introduction

The use of infinite sums goes back in time much further, apparently, than the study of sequences. The sum

$$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\frac{1}{16}+\frac{1}{32}+\frac{1}{64}+\cdots=2$$

has been long known. It is quite easy to convince oneself that this must be valid by arithmetic or geometric "reasoning." After all, just start adding and keeping track of the sum as you progress:

$$1,\ 1\tfrac{1}{2},\ 1\tfrac{3}{4},\ 1\tfrac{7}{8},\ 1\tfrac{15}{16},\ \ldots.$$

Figure 3.1 makes this seem transparent.

But there is a serious problem of meaning here. A finite sum is well defined, an infinite sum is not. Neither humans nor computers can add an infinite column of numbers.

The meaning that is commonly assigned to the preceding sum appears in the following computations:

$$1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots=\lim_{n\to\infty}\left\{1+\frac{1}{2}+\frac{1}{4}+\frac{1}{8}+\cdots+\frac{1}{2^n}\right\}$$

$$=\lim_{n\to\infty}\left\{2\left[1-\frac{1}{2^{n+1}}\right]\right\}=2.$$

This reduces the computation of an infinite sum to that of a finite sum followed by a limit operation. Indeed this is exactly what we were doing when we computed $1,\ 1\frac{1}{2},\ 1\frac{3}{4},\ 1\frac{7}{8},\ 1\frac{15}{16},\ldots$ and felt that this was a compelling reason for thinking of the sum as 2.

In terms of the development of the theory of this textbook this seems entirely natural and hardly surprising. We have mastered sequences in Chapter 2 and now pass to infinite sums in Chapter 3 using the methods of sequences. Historically this was not the case. Infinite summations appear to have been studied and used long before any

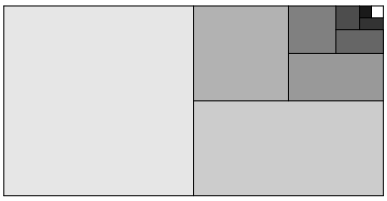


Figure 3.1. $1+1/2+1/4+1/8+1/16+\cdots=2$.

development of sequences and sequence limits. Indeed, even to form the notion of an infinite sum as previously, it would seem that we should already have some concept of sequences, but this is not the way things developed.

It was only by the time of Cauchy that the modern theory of infinite summation was developed using sequence limits as a basis for the theory. We can transfer a great deal of our expertise in sequential limits to the problem of infinite sums. Even so, the study in this chapter has its own character and charm. In many ways infinite sums are much more interesting and important to analysis than sequences.

3.2 Finite Sums

We should begin our discussion of infinite sums with finite sums. There is not much to say about finite sums. Any finite collection of real numbers may be summed in any order and any grouping. That is not to say that we shall not encounter *practical* problems in this. For example, what is the sum of the first 10^{100} prime numbers? No computer or human could find this within the time remaining in this universe. But there is no *mathematical* problem in saying that it is defined; it is a sum of a finite number of real numbers.

There are a number of notations and a number of skills that we shall need to develop in order to succeed at the study of infinite sums that is to come. The notation of such summations may be novel. How best to write out a symbol indicating that some set of numbers

$$\{a_1, a_2, a_3, \ldots, a_n\}$$

has been summed? Certainly

$$a_1 + a_2 + a_3 + \cdots + a_n$$

is too cumbersome a way of writing such sums. The following have proved to communicate much better:

$$\sum_{i\in I} a_i$$

where I is the set $\{1,2,3,\ldots,n\}$ or

$$\sum_{1\le i\le n} a_i \quad \text{or} \quad \sum_{i=1}^{n} a_i.$$

Here the Greek letter Σ, corresponding to an uppercase "S," is used to indicate a sum.

It is to Leonhard Euler (1707–1783) that we owe this sigma notation for sums (first used by him in 1755). The notations $f(x)$ for functions, e and π, i for $\sqrt{-1}$ are also his. These alone indicate the level of influence he has left. In his lifetime he wrote 886 papers and books and is considered the most prolific writer of mathematics that has lived.

The usual rules of elementary arithmetic apply to finite sums. The commutative, associative, and distributive rules assume a different look when written in Euler's notation:

$$\sum_{i\in I} a_i + \sum_{i\in I} b_i = \sum_{i\in I} (a_i + b_i),$$

$$\sum_{i\in I} ca_i = c\sum_{i\in I} a_i,$$

and

$$\left(\sum_{i\in I} a_i\right) \times \left(\sum_{j\in J} b_j\right) = \sum_{i\in I}\left(\sum_{j\in J} a_i b_j\right) = \sum_{j\in J}\left(\sum_{i\in I} a_i b_j\right).$$

Each of these can be checked mainly by determining the meanings and seeing that the notation produces the correct result.

Occasionally in applications of these ideas one would like a simplified expression for a summation. The best known example is perhaps

$$\sum_{k=1}^{n} k = 1+2+3+\cdots+n = \frac{n(n+1)}{2},$$

which is easily proved. When a sum of n terms for a general n has a simpler expression such as this it is usual to say that it has been expressed in *closed form*. Novices, seeing this, usually assume that any summation with some degree of regularity should allow a closed form expression and that it is always important to get a closed form expression. If not, what can you do with a sum that cannot be simplified?

One of the simplest of sums

$$1+\tfrac{1}{2}+\tfrac{1}{3}+\cdots+\tfrac{1}{n} = \textstyle\sum_{k=1}^{n} \frac{1}{k}$$

does not allow any convenient formula, expressing the sum as some simple function of n. This is typical. It is only the rarest of summations that will allow simple formulas. Our work is mostly in *estimating* such expressions; we hardly ever succeed in computing them exactly.

Even so, there are a few special cases that should be remembered and which make our task in some cases much easier.

Telescoping Sums. If a sum can be rewritten in the special form below, a simple computation (canceling s_1, s_2, etc.) gives the following closed form:

$$(s_1 - s_0) + (s_2 - s_1) + (s_3 - s_2) + (s_4 - s_3) + \cdots + (s_n - s_{n-1}) = s_n - s_0.$$

It is convenient to call such a sum "telescoping" as an indication of the method that can be used to compute it.

Example 3.1: For a specific example of a sum that can be handled by considering it as telescoping, consider the sum

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} = \frac{1}{1\cdot 2} + \frac{1}{2\cdot 3} + \frac{1}{3\cdot 4} + \frac{1}{4\cdot 5} + \cdots + \frac{1}{(n-1)\cdot n}.$$

A closed form is available since, using partial fractions, each term can be expressed as

$$\frac{1}{k(k+1)} = \frac{1}{k} - \frac{1}{k+1}.$$

Thus

$$\sum_{k=1}^{n} \frac{1}{k(k+1)} =$$

$$\sum_{k=1}^{n} \left(\frac{1}{k} - \frac{1}{k+1} \right) = 1 - \frac{1}{n+1}.$$

The exercises contain a number of other examples of this type. ◀

Geometric Progressions. If the terms of a sum are in a geometric progression (i.e., if each term is some constant factor times the previous term), then a closed form for any such sum is available:

$$1 + r + r^2 + \cdots + r^{n-1} + r^n = \frac{1 - r^{n+1}}{1 - r}. \tag{1}$$

This assumes that $r \neq 1$; if $r = 1$ the sum is easily seen to be just $n+1$. The formula in (1) can be proved by converting to a telescoping sum. Consider instead $(1-r)$ times the preceding sum:

$$(1-r)(1 + r + r^2 + \cdots + r^{n-1} + r^n) = (1-r) + (r - r^2) + \cdots + (r^n - r^{n+1}).$$

Now add this up as a telescoping sum to obtain the formula stated in (1).

Any geometric progression assumes the form

$$A + Ar + Ar^2 + \cdots + Ar^n = A(1 + r + r^2 + \cdots + r^n)$$

and formula (1) (which should be memorized) is then applied.

Summation By Parts. Sums are frequently given in a form such as

$$\sum_{k=1}^{n} a_k b_k$$

for sequences $\{a_k\}$ and $\{b_k\}$. If a formula happens to be available for

$$s_n = a_1 + a_2 + \cdots + a_n,$$

then there is a frequently useful way of rewriting this sum (using $s_0 = 0$ for convenience):

$$\sum_{k=1}^{n} a_k b_k = \sum_{k=1}^{n} (s_k - s_{k-1}) b_k$$

$$= s_1(b_1 - b_2) + s_2(b_2 - b_3) + \cdots + s_{n-1}(b_{n-1} - b_n) + s_n b_n.$$

Usually some extra knowledge about the sequences $\{s_k\}$ and $\{b_k\}$ can then be used to advantage. The computation is trivial (it is all contained in the preceding equation which is easily checked). Sometimes this summation formula is referred to as Abel's transformation after the Norwegian mathematician Niels Henrik Abel (1802–1829), who was one of the founders of the rigorous theory of infinite sums. It is the analog for finite sums of the integration by parts formula of calculus.

Abel's most important contributions are to analysis but he is forever immortalized in group theory (to which he made a small contribution) by the fact that commutative groups are called "Abelian."

Exercises

3.2.1 Prove the formula

$$\sum_{k=1}^{n} k = \frac{n(n+1)}{2}.$$

3.2.2 Give a formal definition of $\sum_{i\in I} a_i$ for any finite set I and any function $a : I \to \mathbb{R}$ that uses induction on the number of elements of I.

SEE NOTE 36

Your definition should be able to handle the case $I = \emptyset$.

3.2.3 Check the validity of the formulas given in this section for manipulating finite sums. Are there any other formulas you can propose and verify?

3.2.4 Is the formula

$$\sum_{i\in I\cup J} a_i = \sum_{i\in I} a_i + \sum_{i\in J} a_i$$

valid?

SEE NOTE 37

3.2.5 Let $I = \{(i, j) : 1 \le i \le m,\ 1 \le j \le n\}$. Show that

$$\sum_{(i,j)\in I} a_{ij} = \sum_{i=1}^{m}\sum_{j=1}^{n} a_{ij}.$$

3.2.6 Give a formula for the sum of n terms of an arithmetic progression. (An arithmetic progression is a list of numbers, each of which is obtained by adding a fixed constant to the previous one in the list.) For the purposes of infinite sums (our concern in this chapter) such a formula will be of little use. Explain why.

3.2.7 Obtain formulas (or find a source for such formulas) for the sums

$$\sum_{k=1}^{n} k^p = 1^p + 2^p + 3^p + \cdots + n^p$$

of the pth powers of the natural numbers where $p = 1, 2, 3, 4, \ldots$. Again, for the purposes of infinite sums such formulas will be of little use.

3.2.8 Explain the (vague) connection between integration by parts and summation by parts.

SEE NOTE 38

3.2.9 Obtain a formula for $\sum_{k=1}^{n}(-1)^k$.

3.2.10 Obtain a formula for

$$2+2\sqrt{2}+4+4\sqrt{2}+8+8\sqrt{2}+\cdots+2^m.$$

3.2.11 Obtain the formula

$$\sin\theta+\sin 2\theta+\sin 3\theta+\sin 4\theta+\cdots+\sin n\theta=\frac{\cos\theta/2-\cos(2n+1)\theta/2}{2\sin\theta/2}.$$

How should the formula be interpreted if the denominator of the fraction is zero?
SEE NOTE 39

3.2.12 Obtain the formula

$$\cos\theta+\cos 3\theta+\cos 5\theta+\cos 7\theta+\cdots+\cos(2n-1)\theta=\frac{\sin 2n\theta}{2\sin\theta}.$$

3.2.13 If

$$s_n=1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\cdots+(-1)^{n+1}\frac{1}{n}$$

show that $1/2\le s_n\le 1$ for all n.

3.2.14 If

$$s_n=1+\frac{1}{2}+\frac{1}{3}+\frac{1}{4}+\cdots+\frac{1}{n}$$

show that $s_{2^n}\ge 1+n/2$ for all n.

3.2.15 Obtain a closed form for

$$\sum_{k=1}^{n}\frac{1}{k(k+2)(k+4)}.$$

3.2.16 Obtain a closed form for

$$\sum_{k=1}^{n}\frac{\alpha r+\beta}{k(k+1)(k+2)}.$$

3.2.17 Let $\{a_k\}$ and $\{b_k\}$ be sequences with $\{b_k\}$ decreasing and

$$|a_1+a_2+\cdots+a_k|\le K$$

for all k. Show that

$$\left|\sum_{k=1}^{n}a_kb_k\right|\le Kb_1$$

for all n.

3.2.18 If r is the interest rate (e.g., $r=.06$) over a period of years, then

$$P(1+r)^{-1}+P(1+r)^{-2}+\cdots+P(1+r)^{-n}$$

is the present value of an annuity of P dollars paid every year, starting next year and for n years. Give a shorter formula for this. (A perpetuity has nearly the same formula but the payments continue forever. See Exercise 3.4.12.)

3.2.19 Define a finite product (product of a finite set of real numbers) by writing

$$\prod_{k=1}^{n}a_k=a_1a_2a_3\ldots a_n.$$

What elementary properties can you determine for products?

3.2.20 Find a closed form expression for

$$\prod_{k=1}^{n}\frac{k^3-1}{k^3+1}.$$

3.3 Infinite Unordered sums

✂ *Advanced section. May be omitted.*

We now pass to the study of infinite sums. We wish to interpret

$$\sum_{i\in I} a_i$$

for an index set I that is infinite. The study of finite sums involves no analysis, no limits, no ε's, in short none of the processes that are special to analysis. To define and study infinite sums requires many of our skills in analysis.

To begin our study imagine that we are given a collection of numbers a_i indexed over an infinite set I (i.e., there is a function $a : I \to \mathbb{R}$) and we wish the sum of the totality of these numbers. If the set I has some structure, then we can use that structure to decide how to start adding the numbers. For example, if a is a sequence so that $I = \mathbb{N}$, then we should likely start adding at the beginning of the sequence:

$$a_1,\ a_1+a_2,\ a_1+a_2+a_3,\ a_1+a_2+a_3+a_4,\ldots$$

and so defining the sum as the limit of this sequence of partial sums.

Another set I would suggest a different order. For example, if $I = \mathbb{Z}$ (the set of all integers), then a popular method of adding these up would be to start off:

$$a_0,\ a_{-1}+a_0+a_1,$$

$$a_{-2}+a_{-1}+a_0+a_1+a_2,$$

$$a_{-3}+a_{-2}+a_{-1}+a_0+a_1+a_2+a_3,\ \ldots$$

once again defining the sum as the limit of this sequence.

It seems that the method of summation and hence defining the meaning of the expression

$$\sum_{i\in I} a_i$$

for infinite sets I must depend on the nature of the set I and hence on the particular problems of the subject one is studying. This is true to some extent. But it does not stop us from inventing a method that will apply to *all* infinite sets I. We must make a definition that takes account of no extra structure or ordering for the set I and just treats it as a set. This is called the unordered sum and the notation $\sum_{i\in I} a_i$ is always meant to indicate that an unordered sum is being considered. The key is just how to pass from finite sums to infinite sums. Both of the previous examples used the idea of taking some finite sums (in a systematic way) and then passing to a limit.

Definition 3.2: Let I be an infinite set and a a function $a : I \to \mathbb{R}$. Then we write

$$\sum_{i\in I} a_i = c$$

and say that the sum *converges* if for every $\varepsilon > 0$ there is a finite set $I_0 \subset I$ so that, for every finite set J, $I_0 \subset J \subset I$,

$$\left|\sum_{i\in J} a_i - c\right| < \varepsilon.$$

A sum that does not converge is said to *diverge*.

Note that we never form a sum of infinitely many terms. The definition always computes finite sums.

Example 3.3: Let us show, directly from the definition, that

$$\sum_{i\in\mathbb{Z}} 2^{-|i|} = 3.$$

If we first sum

$$\sum_{-N\le i\le N} 2^{-|i|}$$

by rearranging the terms into the sum

$$1+2(2^{-1}+2^{-2}+\cdots+2^{-N})$$

we can see why the sum is likely to be 3. Let $\varepsilon > 0$ and choose N so that $2^{-N} < \varepsilon/4$. From the formula for a finite geometric progression we have

$$\left|\sum_{-N\le i\le N} 2^{-|i|} - 3\right| = 2|(2^{-1}+2^{-2}+\cdots+2^{-N})-1| = 2(2^{-N}) < \varepsilon/2.$$

Also, if $K \subset \mathbb{Z}$ with K finite and $|k| > N$ for all $k \in K$, then

$$\sum_{k\in K} 2^{-|k|} < 2(2^{-N}) < \varepsilon/2$$

again from the formula for a finite geometric progression. Let

$$I_0 = \{i \in \mathbb{Z} : -N \le i \le N\}.$$

If $I_0 \subset J \subset \mathbb{Z}$ with J finite then

$$\left|\sum_{i\in J} 2^{-|i|} - 3\right| = \left|\sum_{-N\le i\le N} 2^{-|i|} - 3\right| + \sum_{i\in J\setminus I_0} 2^{-|i|} < \varepsilon$$

as required. ◀

3.3.1 Cauchy Criterion

✂ *Advanced section. May be omitted.*

In most theories of convergence one asks for a necessary and sufficient condition for convergence. We saw in studying sequences that the Cauchy criterion provided such a condition for the convergence of a sequence. There is usually in any theory of this kind a type of Cauchy criterion. Here is the Cauchy criterion for sums.

Theorem 3.4: *A necessary and sufficient condition that the sum $\sum_{i\in I} a_i$ converges is that for every $\varepsilon > 0$ there is a finite set I_0 so that*

$$\left|\sum_{i\in J} a_i\right| < \varepsilon$$

for every finite set $J \subset I$ that contains no elements of I_0 (i.e., for all finite sets $J \subset I \setminus I_0$).

Proof. As usual in Cauchy criterion proofs, one direction is easy to prove. Suppose that $\sum_{i\in I} a_i = C$ converges. Then for every $\varepsilon > 0$ there is a finite set I_0 so that

$$\left|\sum_{i\in K} a_i - C\right| < \varepsilon/2$$

for every finite set $I_0 \subset K \subset I$. Let J be a finite subset of $I \setminus I_0$ and consider taking a sum over $K = I_0 \cup J$. Then

$$\left|\sum_{i\in I_0\cup J} a_i - C\right| < \varepsilon/2$$

and

$$\left|\sum_{i\in I_0} a_i - C\right| < \varepsilon/2.$$

By subtracting these two inequalities and remembering that

$$\sum_{i\in I_0\cup J} a_i = \sum_{i\in J} a_i + \sum_{i\in I_0} a_i$$

(since I_0 and J are disjoint) we obtain

$$\left|\sum_{i\in J} a_i\right| < \varepsilon.$$

This is exactly the Cauchy criterion.

Conversely, suppose that the sum does satisfy the Cauchy criterion. Then, applying that criterion to $\varepsilon = 1, 1/2, 1/3, \ldots$ we can choose a sequence of finite sets $\{I_n\}$ so that

$$\left|\sum_{i\in J} a_i\right| < 1/n$$

for every finite set $J \subset I \setminus I_n$. We can arrange our choices to make

$$I_1 \subset I_2 \subset I_3 \subset \ldots$$

so that the sequence of sets is increasing.

Let

$$c_n = \sum_{i\in I_n} a_i$$

Then for any $m > n$,

$$|c_n - c_m| = \left|\sum_{i\in I_m\setminus I_n} a_i\right| < 1/n.$$

It follows from this that $\{c_n\}$ is a Cauchy sequence of real numbers and hence converges to some real number c. Let $\varepsilon > 0$ and choose an integer N larger than $2/\varepsilon$ and so that $|c_N - c| < \varepsilon/2$. Then, for any $n > N$ and any finite set J with $I_N \subset J \subset I$,

$$\left|\sum_{i\in J} a_i - c\right| \le \left|\sum_{i\in I_N} a_i - c_N\right| + |c_N - c| + \left|\sum_{i\in J\setminus I_N} a_i\right| < 0 + \varepsilon/2 + 1/N < \varepsilon.$$

By definition, then,

$$\sum_{i\in I} a_i = c$$

and the theorem is proved. ■

All But Countably Many Terms in a Convergent Sum Are Zero. Our next theorem shows that having "too many" numbers to add up causes problems. If the set I is not countable then most of the a_i that we are to add up should be zero if the sum is to exist. This shows too that the theory of sums is in an essential way limited to taking sums over countable sets. It is notationally possible to have a sum

$$\sum_{x\in[0,1]} f(x)$$

but that sum cannot be defined unless $f(x)$ is mostly zero with only countably many exceptions.

Theorem 3.5: *Suppose that $\sum_{i\in I} a_i$ converges. Then $a_i = 0$ for all $i \in I$ except for a countable subset of I.*

Proof. We shall use Exercise 3.3.2, where it is proved that for any convergent sum there is a positive integer M so that all the sums

$$\left|\sum_{i\in I_0} a_i\right| \leq M$$

for any finite set $I_0 \subset I$. Let m be an integer. We ask how many elements a_i are there such that $a_i > 1/m$? It is easy to see that there are at most Mm of them since if there were any more our sum would exceed M. Similarly, there are at most Mm terms such that $-a_i > 1/m$. Thus each element of $\{a_i : i \in I\}$ that is not zero can be given a "rank" m depending on whether

$$1/m < a_i \leq 1/(m-1) \text{ or } 1/m < -a_i \leq 1/(m-1).$$

As there are only finitely many elements at each rank, this gives us a method for listing all of the nonzero elements in $\{a_i : i \in I\}$ and so this set is countable. ■

The elementary properties of unordered sums are developed in the exercises. These sums play a small role in analysis, a much smaller role than the ordered sums we shall consider in the next sections. The methods of proof, however, are well worth studying since they are used in some form or other in many parts of analysis. These exercises offer an interesting setting in which to test your skills in analysis, skills that will play a role in all of your subsequent study.

Exercises

3.3.1 Show that if $\sum_{i\in I} a_i$ converges, then the sum is unique.

SEE NOTE 40

3.3.2 Show that if $\sum_{i\in I} a_i$ converges, then there is a positive number M so that all the sums

$$\left|\sum_{i\in I_0} a_i\right| \leq M$$

for any finite set $I_0 \subset I$.

SEE NOTE 41

3.3.3 Suppose that all the terms in the sum $\sum_{i\in I} a_i$ are nonnegative and that there is a positive number M so that all the sums

$$\sum_{i\in I_0} a_i \leq M$$

for any finite set $I_0 \subset I$. Show that $\sum_{i\in I} a_i$ must converge.

SEE NOTE 42

3.3.4 Show that if $\sum_{i\in I} a_i$ converges so too does $\sum_{i\in J} a_i$ for every subset $J \subset I$.

3.3.5 Show that if $\sum_{i\in I} a_i$ converges and each $a_i \geq 0$, then

$$\sum_{i\in I} a_i = \sup\left\{\sum_{i\in J} a_i : \ J \subset I, J \text{ finite}\right\}.$$

3.3.6 Each of the rules for manipulation of the finite sums of Section 3.2 can be considered for infinite unordered sums. Formulate the correct statement and prove what you think to be the analog of these statements that we know hold for finite sums:

$$\sum_{i\in I} a_i + \sum_{i\in I} b_i = \sum_{i\in I} (a_i + b_i)$$

$$\sum_{i\in I} ca_i = c\sum_{i\in I} a_i$$

$$\sum_{i\in I} a_i \times \sum_{i\in J} b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j = \sum_{j\in J}\sum_{i\in I} a_i b_j.$$

3.3.7 Prove that

$$\sum_{i\in I\cup J} a_i + \sum_{i\in I\cap J} a_i = \sum_{i\in I} a_i + \sum_{i\in J} a_i$$

under appropriate convergence assumptions.

3.3.8 Let $\sigma : I \to J$ one-to-one and onto. Establish that

$$\sum_{j\in J} a_j = \sum_{i\in I} a_{\sigma(i)}$$

under appropriate convergence assumptions.

3.3.9 Find the sum

$$\sum_{i\in \mathbb{N}} \frac{1}{2^i}.$$

SEE NOTE 43

3.3.10 Show that

$$\sum_{i\in \mathbb{N}} \frac{1}{i}$$

diverges. Are there any infinite subsets $J \subset \mathbb{N}$ such that

$$\sum_{i\in J} \frac{1}{i}$$

converges?

3.3.11 Show that $\sum_{i\in I} a_i$ converges if and only if both $\sum_{i\in I}[a_i]^+$ and $\sum_{i\in I}[a_i]^-$ converge and that

$$\sum_{i\in I} a_i = \sum_{i\in I}[a_i]^+ - \sum_{i\in I}[a_i]^-$$

and

$$\sum_{i\in I} |a_i| = \sum_{i\in I}[a_i]^+ + \sum_{i\in I}[a_i]^-.$$

See Note 44

3.3.12 Compute

$$\sum_{(i,j)\in\mathbb{N}\times\mathbb{N}} 2^{-i-j}.$$

What kind of *ordered* sum would seem natural here (in the way that ordered sums over $\mathbb{N}$ and $\mathbb{Z}$ were considered in this section)?

See Note 45

3.4 Ordered Sums: Series

For the vast majority of applications, one wishes to sum not an arbitrary collection of numbers but most commonly some sequence of numbers:

$$a_1 + a_2 + a_3 + \dots.$$

The set $\mathbb{N}$ of natural numbers has an order structure, and it is not in our best interests to ignore that order since that is the order in which the sequence is presented to us.

The most compelling way to add up a sequence of numbers is to begin accumulating:

$$a_1,\ a_1 + a_2,\ a_1 + a_2 + a_3,\ a_1 + a_2 + a_3 + a_4, \dots$$

and to define the sum as the limit of this sequence. This is what we shall do.

If you studied Section 3.3 on unordered summation you should also compare this "ordered" method with the unordered method. The ordered sum of a sequence is called a *series* and the notation

$$\sum_{k=1}^{\infty} a_k$$

is used exclusively for this notion.

Definition 3.6: Let $\{a_k\}$ be a sequence of real numbers. Then we write

$$\sum_{k=1}^{\infty} a_k = c$$

and say that the series *converges* if the sequence

$$s_n = \sum_{k=1}^{n} a_k$$

(called the *sequence of partial sums of the series*) converges to c. If the series does not converge it is said to be *divergent*.

This definition reduces the study of series to the study of sequences. We already have a highly developed theory of convergent sequences in Chapter 2 that we can apply to develop a theory of series. Thus we can rapidly produce a fairly deep theory of series from what we already know. As the theory develops, however, we shall see that it begins to take a character of its own and stops looking like a mere application of sequence ideas.

3.4.1 Properties

The following short harvest of theorems we obtain directly from our sequence theory. The convergence or divergence of a series $\sum_{k=1}^{\infty} a_k$ depends on the convergence or divergence of the sequence of partial sums

$$s_n = \sum_{k=1}^{n} a_k$$

and the value of the series is the limit of the sequence. To prove each of the theorems we now list requires only to find the correct theorem on sequences from Chapter 2. This is left as Exercise 3.4.2.

Theorem 3.7: *If a series $\sum_{k=1}^{\infty} a_k$ converges, then the sum is unique.*

Theorem 3.8: *If both series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ converge, then so too does the series*

$$\sum_{k=1}^{\infty} (a_k + b_k)$$

and

$$\sum_{k=1}^{\infty} (a_k + b_k) = \sum_{k=1}^{\infty} a_k + \sum_{k=1}^{\infty} b_k.$$

Theorem 3.9: *If the series $\sum_{k=1}^{\infty} a_k$ converges, then so too does the series $\sum_{k=1}^{\infty} ca_k$ for any real number c and*

$$\sum_{k=1}^{\infty} ca_k = c \sum_{k=1}^{\infty} a_k.$$

Theorem 3.10: *If both series $\sum_{k=1}^{\infty} a_k$ and $\sum_{k=1}^{\infty} b_k$ converge and $a_k \leq b_k$ for each k, then*

$$\sum_{k=1}^{\infty} a_k \leq \sum_{k=1}^{\infty} b_k.$$

Theorem 3.11: *Let $M \geq 1$ be any integer. Then the series*

$$\sum_{k=1}^{\infty} a_k = a_1 + a_2 + a_3 + a_4 + \ldots$$

converges if and only if the series

$$\sum_{k=1}^{\infty} a_{M+k} = a_{M+1} + a_{M+2} + a_{M+3} + a_{M+4} + \ldots$$

converges.